

Power Electronics

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Desk D216

Layout

- Non isolated choppers
 - Introduction
 - Direct conversion
 - Buck
 - Boost
 - Reversible choppers
 - Half bridge
 - H – bridge
 - Inverter
 - Energy storage based chopper
 - Buck-boost
 - Cuck
 - SEPIC
- Isolated power supplies
- Power components



Progress

- Lesson : 8 x 2h
- Tutorials : 9 x 2h
- Labs : 4 x 4h
 - Buck chopper
 - Reversible chopper for DC motor
 - Switch mode power supply Flyback
 - Switch mode power supply Forward
- Marks : Final exam - Labs



Prerequisites

- Kirchhoff's circuit laws
- Average value of a signal, integral calculus, RMS value
- AC and DC real power, power factor, complex representation, phasor
- Transformers (magnetic field, flux)
- Linear control (Laplace transform, Bode diagram, gain/phase margin, PI controller)



Introduction

Direct conversion

Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper



Introduction

Introduction

Direct conversion

Reversible
choppers

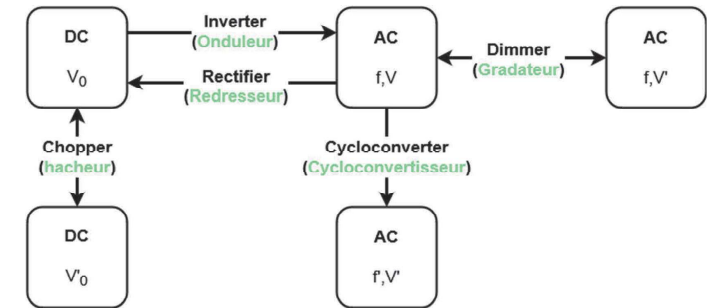
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Closed loop
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Energy conversion: principles



Introduction

Direct conversion

Reversible
choppers

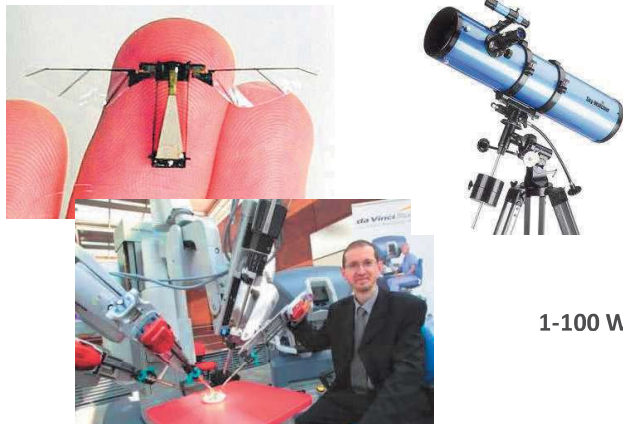
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Micro-mechanisms



1-100 W

Introduction

Direct conversion

Reversible
choppers

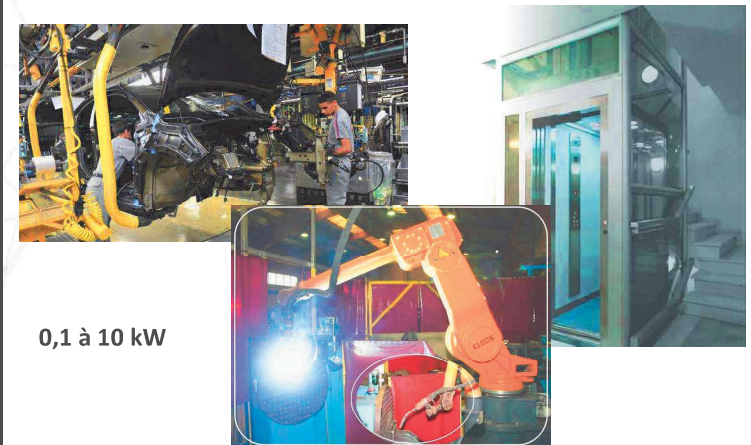
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Closed loop
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based chopper



Use : automatism



0,1 à 10 kW

Introduction

Direct conversion

Reversible
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Use : rail traction



100 kW

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High speed rail traction



10 MW

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Photovoltaic plant



300 MWc

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Direct conversion

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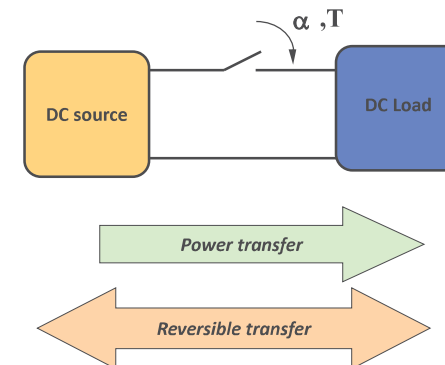
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«Chopper » function : a switching electronic



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Introduction

Direct conversion

Reversible
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Link power sources

Voltage Source

Current Source

Active
(Energy generator)

Passive
(Energy receiver)

Theorem
Only two sources with different category
can be directly connected.

Introduction

Direct conversion

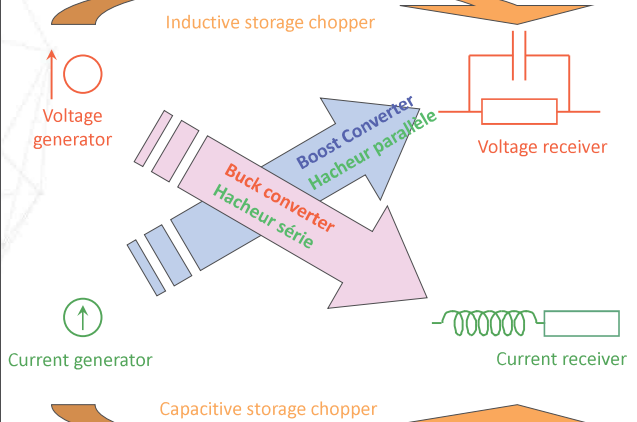
Reversible
choppers

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based chopper

Non reversible chopper



Introduction

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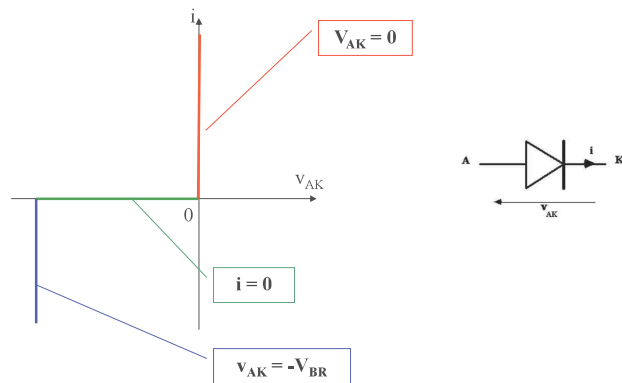
Reversible
choppers

Inverters

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Energy storage
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« Perfect » diode



Introduction

Direct conversion

Reversible
choppers

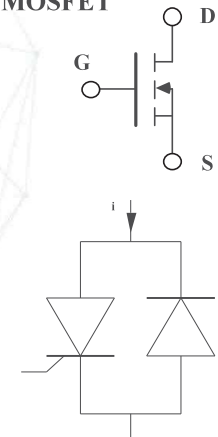
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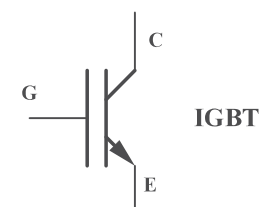
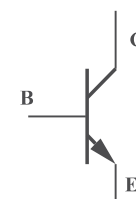
Energy storage
based chopper

Controlled switch

MOSFET



Bipolar
transistor



Introduction

Direct conversion

Reversible choppers

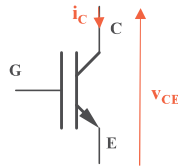
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Closed loop chopper control

Energy storage based chopper



« Perfect » controlled switch ex : IGBT



- On state: « short circuit »
 - $v_{CE} = 0$, i_C fixed by circuit
- Off state: « open circuit »
 - $i_C = 0$, v_{CE} fixed by circuit

Introduction

Direct conversion

Reversible choppers

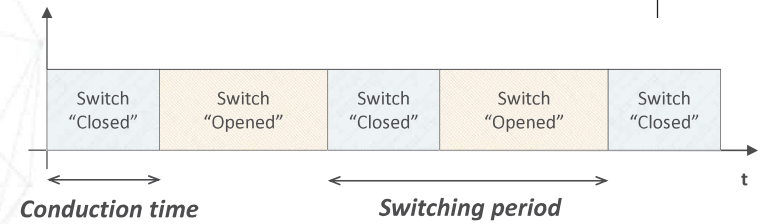
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Closed loop chopper control

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Duty cycle



$$\alpha = \frac{\text{Conduction time}}{\text{Switching period}}$$

Introduction

Direct conversion

Buck chopper

Boost chopper

Reversible choppers

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Energy storage based chopper



Buck Chopper

Introduction

Direct conversion

Buck chopper

Boost chopper

Reversible choppers

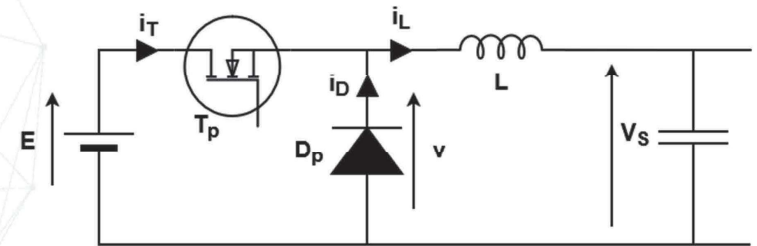
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Energy storage based chopper



Buck chopper : principle



Hypothesis : $V_s = \text{constant}$

Introduction

Direct conversion

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Boost chopper

Reversible choppers

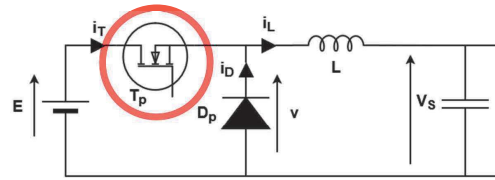
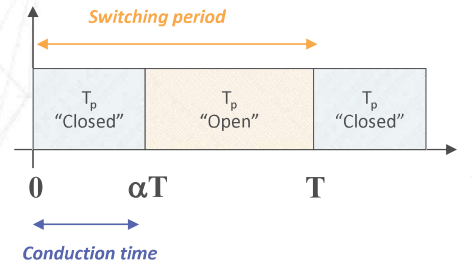
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Control



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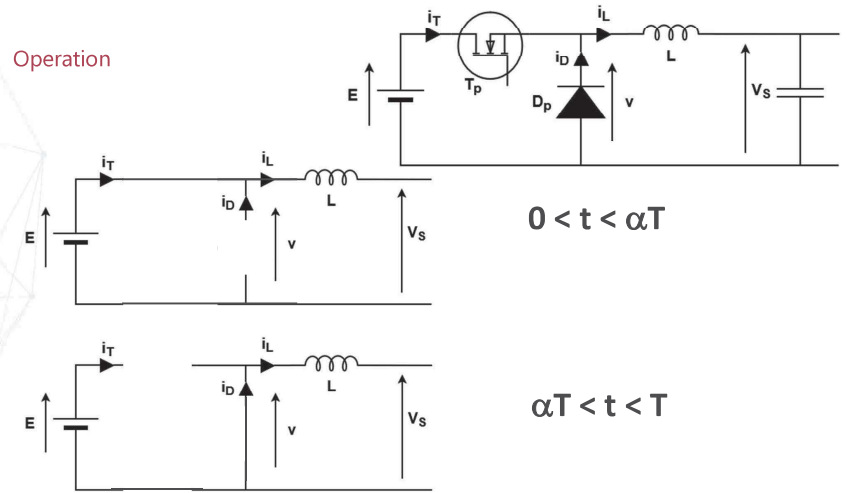
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Operation



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Reversible choppers

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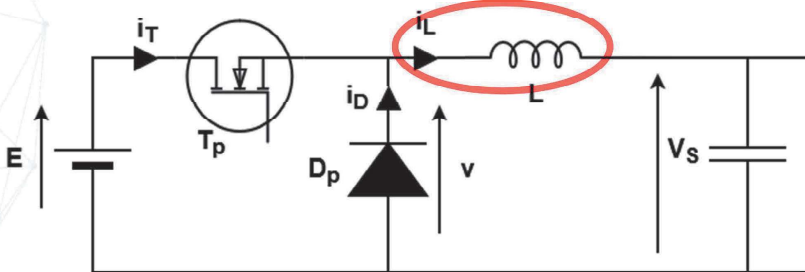
Closed loop chopper control

Energy storage based chopper



Modes

- **Continuous conduction mode:** $i_L > 0$



- **Discontinuous conduction mode:** $i_L(t_0) = 0 ; t_0 < T$

Introduction

Direct conversion

Buck chopper

Boost chopper

Reversible choppers

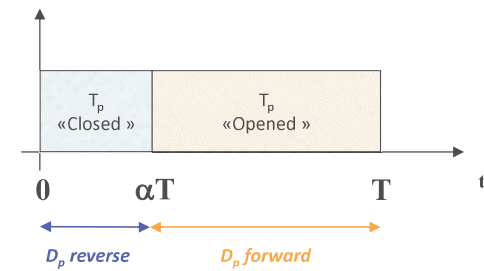
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Continuous mode



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Reversible
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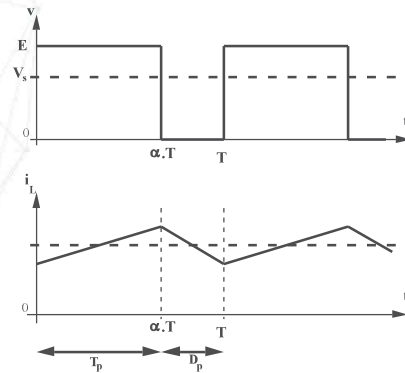
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Closed loop
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Energy storage
based chopper



Continuous mode: Chronograms



$$\bullet \quad 0 < t < \alpha T$$

$$\frac{di}{dt} = \frac{E - V_s}{L}$$

$$\bullet \quad \alpha T < t < T$$

$$\frac{di}{dt} = -\frac{V_s}{L}$$

$$E > V_s$$

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Direct conversion

Buck chopper
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Reversible
choppers

Inverters

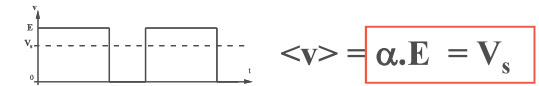
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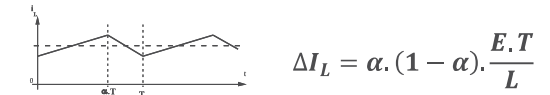


Continuous mode (Calculation)

Output voltage

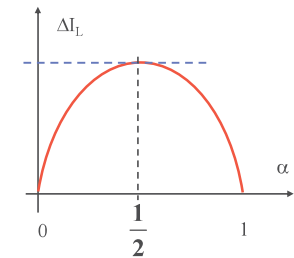


Current ripple i_L



Maximum ripple

$$\Delta I_{L \max} = \frac{E \cdot T}{4 L}$$



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Introduction

Direct conversion

Buck chopper
Boost chopper

Reversible
choppers

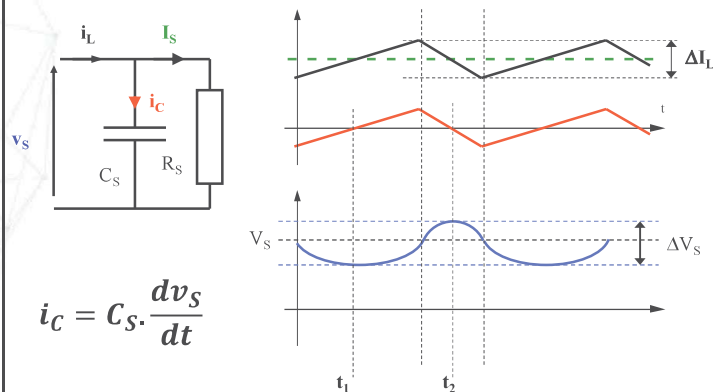
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Energy storage
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Continuous mode: Output filtering principle



$$i_C = C_S \cdot \frac{dv_s}{dt}$$

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Direct conversion

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Reversible
choppers

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Continuous mode: Output filtering calculation

$$t_1 = \alpha \frac{T}{2}$$

$$t_2 = \frac{(1+\alpha) \cdot T}{2}$$

$$\Delta I_L = \alpha \cdot (1 - \alpha) \cdot \frac{E \cdot T}{L}$$

$$\Delta V_S = \frac{1}{C_S} \cdot \int_{t_1}^{t_2} i_C(t) \cdot dt = \frac{1}{C_S} \cdot \frac{1}{2} \cdot \frac{\Delta I_L}{2} \cdot (t_2 - t_1) = \frac{\Delta I_L \cdot T}{8 \cdot C_S}$$

$$C_S = \frac{\alpha \cdot (1 - \alpha) \cdot E \cdot T^2}{8 \cdot L \cdot \Delta V_S}$$

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Direct conversion

Buck chopper
Boost chopper

Reversible
choppers

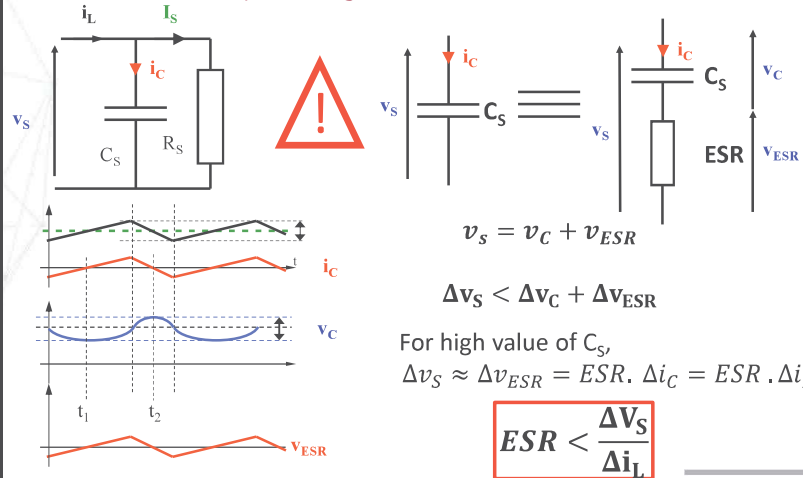
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Energy storage
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Continuous mode: Output filtering calculation



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Direct conversion

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Boost chopper

Reversible
choppers

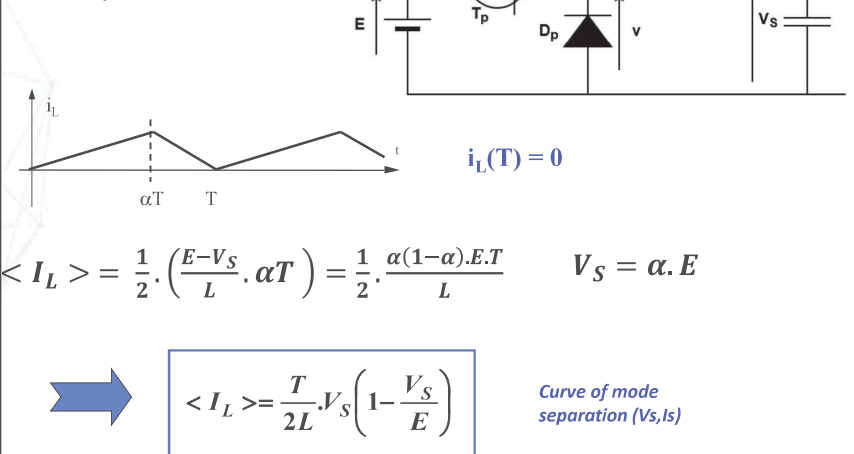
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Boundary mode



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Direct conversion

Buck chopper
Boost chopper

Reversible
choppers

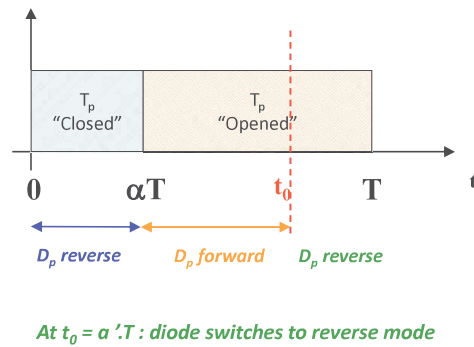
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Discontinuous mode



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Boost chopper

Reversible
choppers

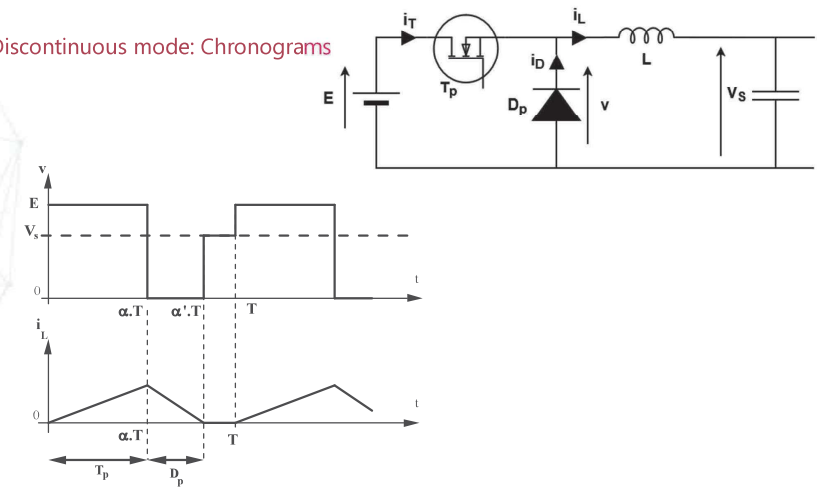
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Discontinuous mode: Chronograms



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Boost chopper

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Closed loop

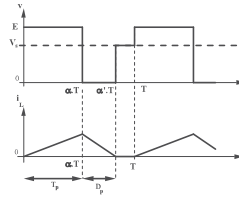
chopper control

Energy storage

based chopper



Discontinuous mode: calculation



$$I_{max} = \frac{E - V_S}{L} \cdot \alpha T = \frac{V_S}{L} \cdot (\alpha' T - \alpha T) \Rightarrow \frac{\alpha}{\alpha'} = \frac{V_S}{E}$$

$$\langle I_L \rangle = \frac{1}{2} \cdot I_{max} \cdot \alpha' = \frac{E - V_S}{2L} \cdot \alpha \cdot \alpha' \cdot T = \frac{\alpha \cdot E \cdot T}{2L} (\alpha' - \alpha)$$

$$\Rightarrow \alpha' = \alpha + \frac{2L \cdot \langle I_L \rangle}{\alpha \cdot E \cdot T}$$

$$V_S = \frac{\alpha \cdot E}{\alpha + \frac{2L \cdot \langle I_L \rangle}{\alpha \cdot E \cdot T}} \quad \text{Output current: } \langle I_L \rangle = I_S$$

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Direct conversion

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Closed loop

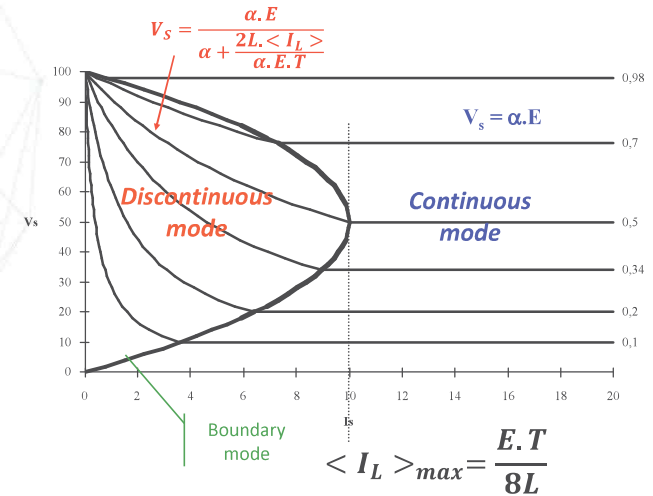
chopper control

Energy storage

based chopper



Output characteristics



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Introduction

Direct conversion

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Boost chopper

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Closed loop

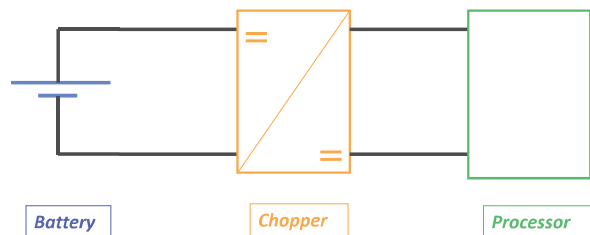
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Example



=> DC supply from battery

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Introduction

Direct conversion

Buck chopper

Boost chopper

Reversible

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Closed loop

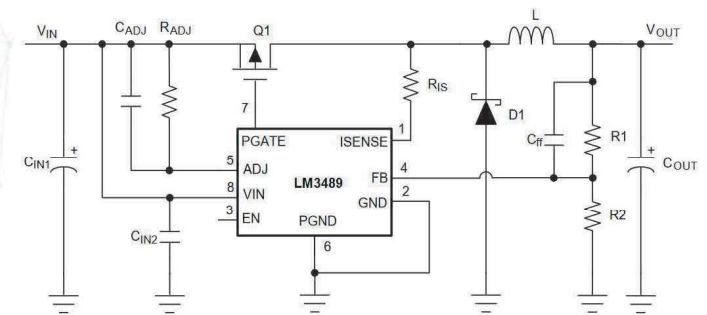
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Example : LM3489



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Direct conversion

Buck chopper

Boost chopper

Reversible

choppers

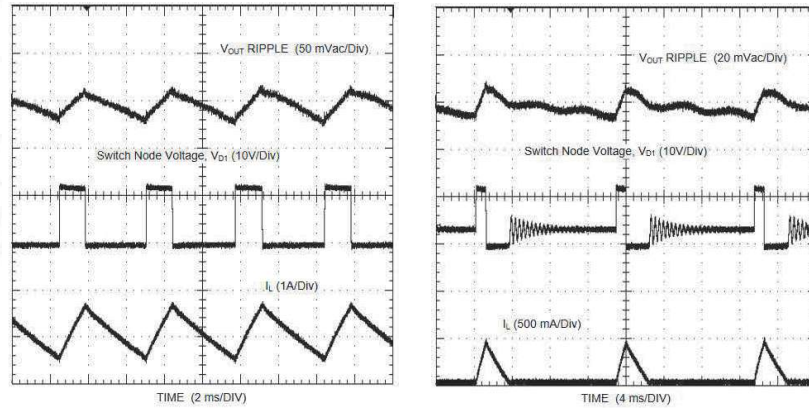
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Example : LM3489



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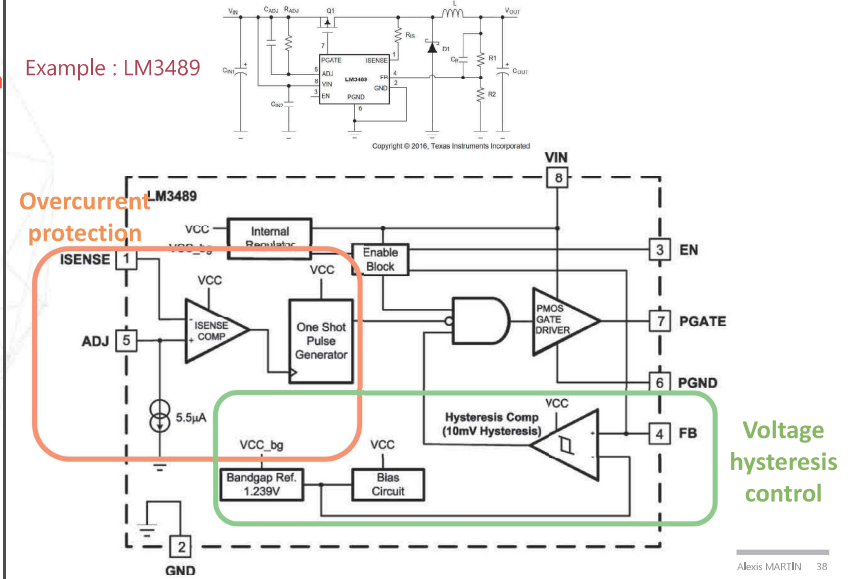
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Example : LM3489



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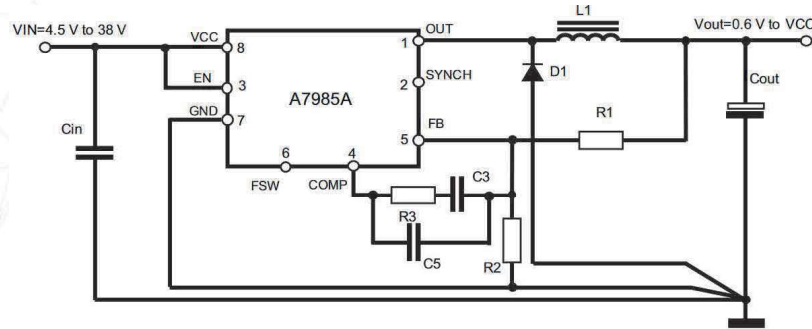
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Example : A7985A



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Direct conversion

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Reversible

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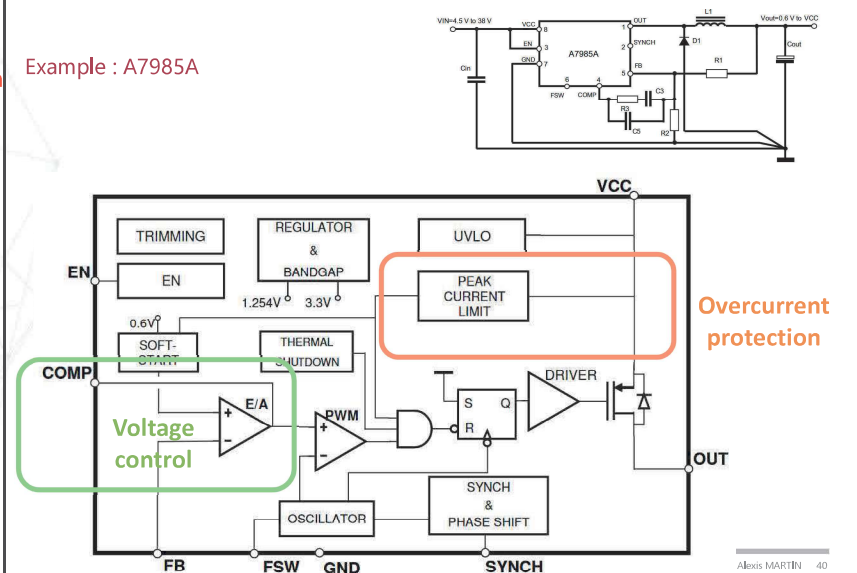
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Example : A7985A



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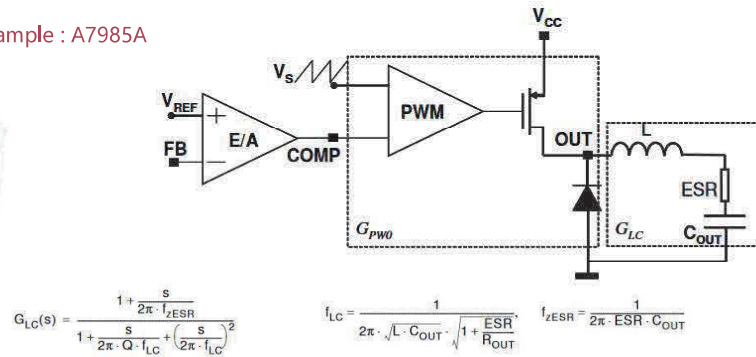
chopper control

Energy storage

based chopper



Example : A7985A



$$G_{LC}(s) = \frac{1 + \frac{s}{2\pi \cdot f_{ZESR}}}{1 + \frac{s}{2\pi \cdot Q \cdot f_{LC}} + \left(\frac{s}{2\pi \cdot f_{LC}}\right)^2}$$

$$f_{LC} = \frac{1}{2\pi \cdot \sqrt{L \cdot C_{OUT}} \cdot \sqrt{1 + \frac{ESR}{R_{OUT}}}}, \quad f_{ZESR} = \frac{1}{2\pi \cdot ESR \cdot C_{OUT}}$$

$$Q = \frac{\sqrt{R_{OUT} \cdot L \cdot C_{OUT} \cdot (R_{OUT} + ESR)}}{L + C_{OUT} \cdot R_{OUT} \cdot ESR}$$

Usually : $f_{LC} < f_{ZESR}$

→ Slightly depends on the load

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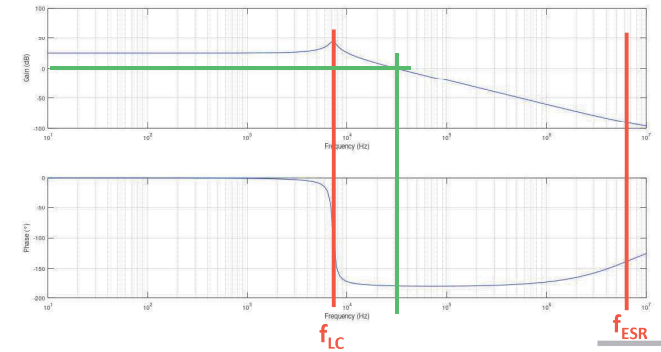
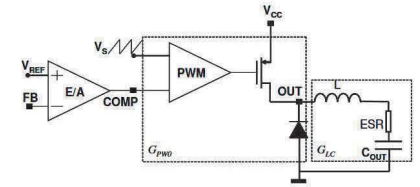
chopper control

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Example : A7985A



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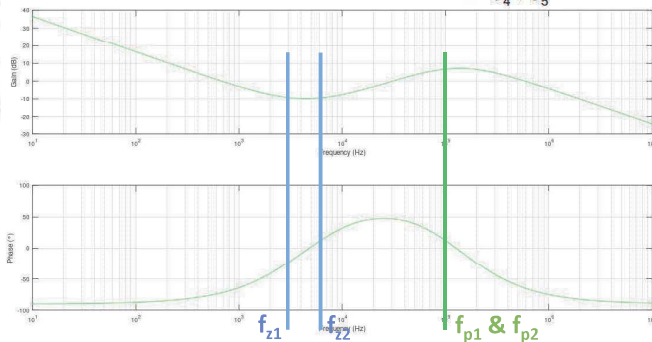


Example : A7985A

If $f_{ZESR} \gg BW$ (desired)
($ESR < 1m\Omega$)

$$\text{Zeros: } f_{Z1} = \frac{1}{2\pi \cdot C_3 \cdot (R_1 + R_3)}, \quad f_{Z2} = \frac{1}{2\pi \cdot R_4 \cdot C_4}$$

$$\text{Poles: } f_{P0} = 0, \quad f_{P1} = \frac{1}{2\pi \cdot R_3 \cdot C_3}, \quad f_{P2} = \frac{1}{2\pi \cdot R_4 \cdot \frac{C_4 \cdot C_5}{C_4 + C_5}}$$



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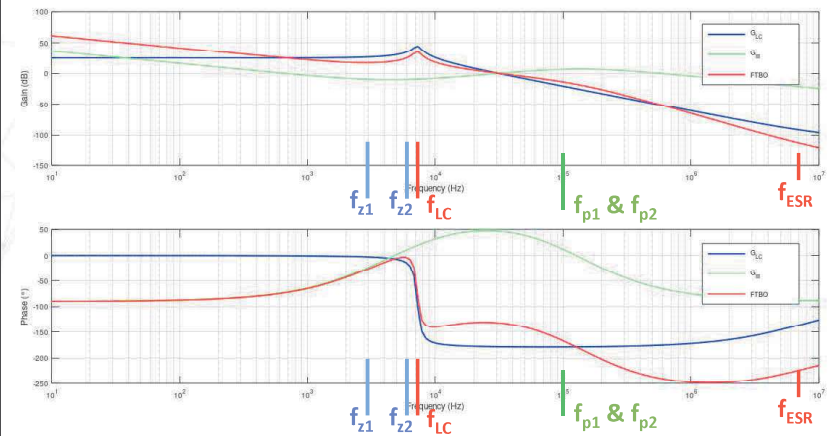
chopper control

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Example : A7985A



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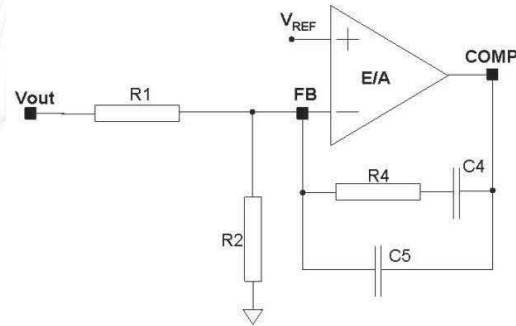
Energy storage based chopper



Example : A7985A

If $f_{zESR} \approx BW$ (desired)
ESR > 30 mΩ

→ Use of the zero of ESR to stabilize the loop



Zero : $f_{z1} = \frac{1}{2\pi \cdot R_4 \cdot C_4}$

Poles : $f_{p0} = 0,$

$$f_{p1} = \frac{1}{2\pi \cdot R_4 \cdot \frac{C_4 \cdot C_5}{C_4 + C_5}}$$

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Boost Chopper

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Boost chopper

Buck chopper : Voltage source



Current receiver

Boost chopper : Current source



Voltage receiver

Current source ⇌ Voltage source + inductor



Voltage receiver ⇌ Battery or capacitive receiver

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Direct conversion

Buck chopper
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Reversible choppers

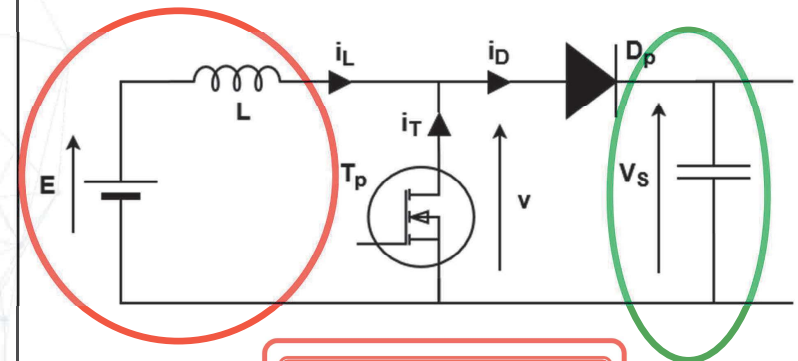
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Boost chopper: principle



Hypothesis : $V_s = \text{constant}$

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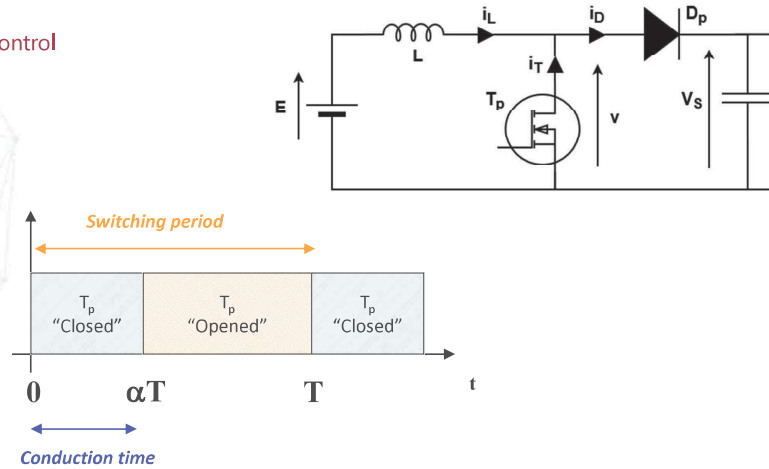
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Control



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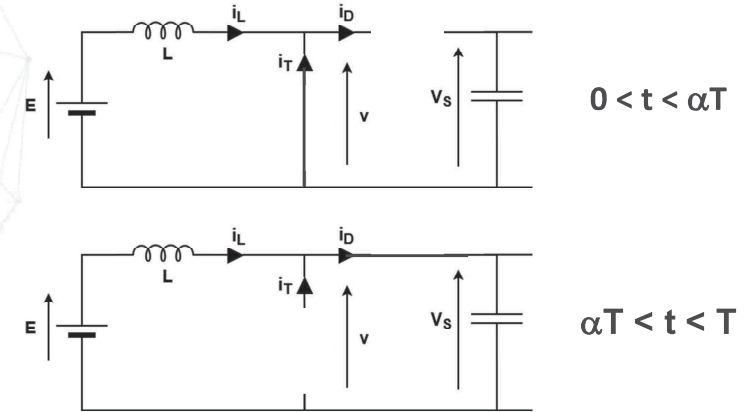
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Principle



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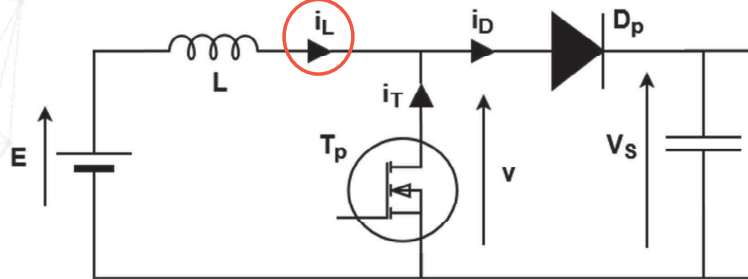
Closed loop
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Modes

- Continuous conduction: $i_L > 0$



- Discontinuous conduction: $i_L(t_0) = 0$; $t_0 < T$

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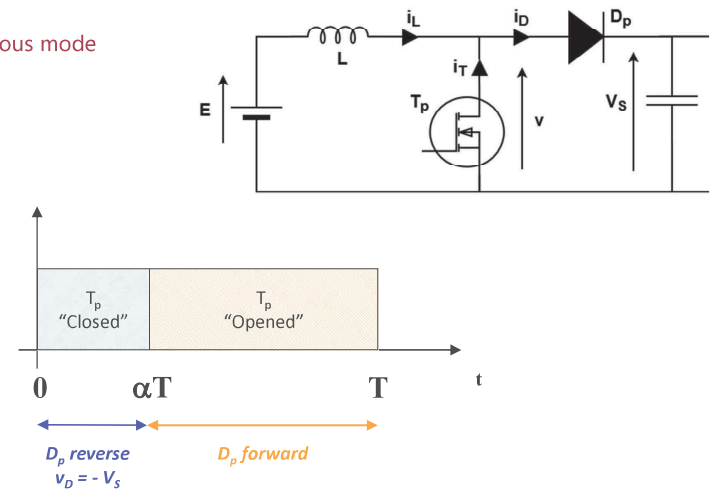
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Continuous mode



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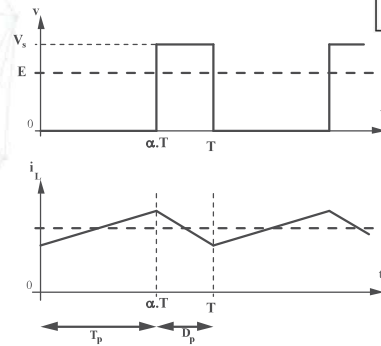
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Continuous mode: Chronograms



$$E < V_s$$

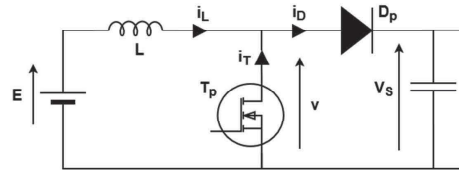
BOOST

$$\bullet \quad 0 < t < \alpha T$$

$$\frac{di}{dt} = \frac{E}{L}$$

$$\bullet \quad \alpha T < t < T$$

$$\frac{di}{dt} = \frac{E - V_s}{L}$$



Introduction

Direct conversion

Buck chopper
Boost chopper

Reversible
choppers

Inverters

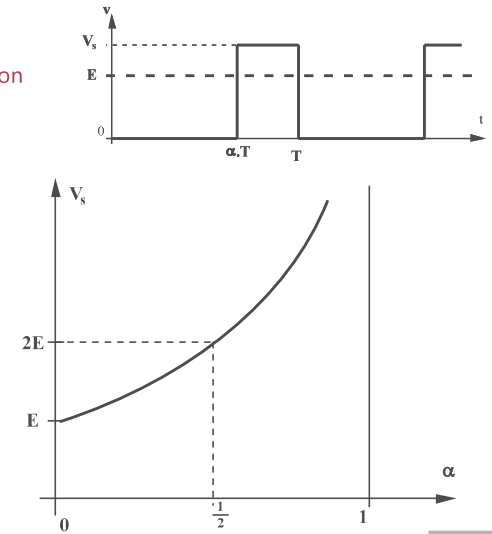
Closed loop
chopper control

Energy storage
based chopper



Continuous mode:
Output voltage calculation

$$V_s = \frac{E}{1 - \alpha}$$



Introduction

Direct conversion

Buck chopper
Boost chopper

Reversible
choppers

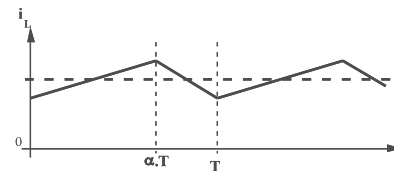
Inverters

Closed loop
chopper control

Energy storage
based chopper



Continuous mode:
Current ripple calculation



Current ripple i_L

$$\Delta I_L = \alpha \cdot \frac{E \cdot T}{L}$$

Maximum ripple ($\alpha = 1$)

$$\Delta I_{L \max} = \frac{E \cdot T}{L}$$

Introduction

Direct conversion

Buck chopper
Boost chopper

Reversible
choppers

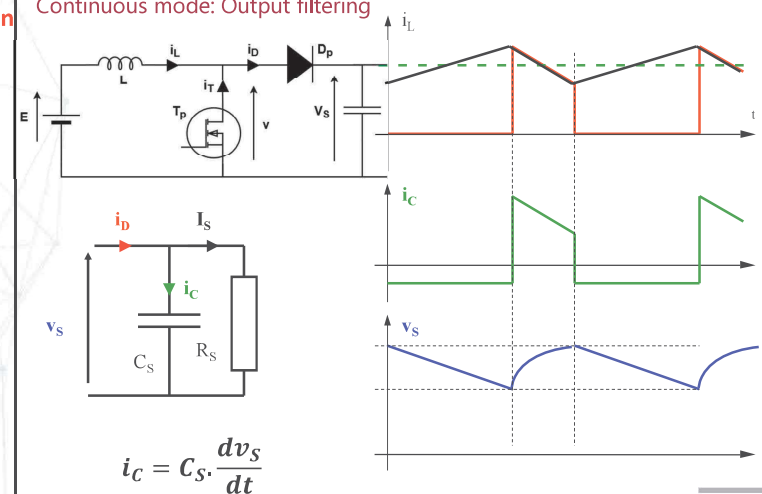
Inverters

Closed loop
chopper control

Energy storage
based chopper



Continuous mode: Output filtering



$$i_c = C_s \cdot \frac{dv_s}{dt}$$

Introduction

Direct conversion

Buck chopper
Boost chopper

Reversible choppers

Inverters

Closed loop chopper control

Energy storage based chopper



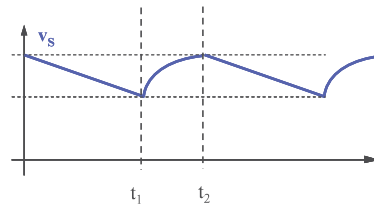
Continuous mode: Output filtering

$$\Delta V_S = \frac{1}{C_S} \cdot \int_{t_1}^{t_2} i_C(t) \cdot dt = \frac{I_S \cdot \alpha T}{C_S}$$

$$\text{or } I_S = \frac{V_S}{R_S} = \frac{E}{(1-\alpha)R_S}$$

$$\Delta V_S = \frac{ET}{R_S C_S} \cdot \frac{\alpha}{1-\alpha}$$

$$C_S = \frac{\alpha \cdot E \cdot T}{(1-\alpha) \cdot R_S \cdot \Delta V_S}$$



Alexis MARTIN 57

Introduction

Direct conversion

Buck chopper
Boost chopper

Reversible choppers

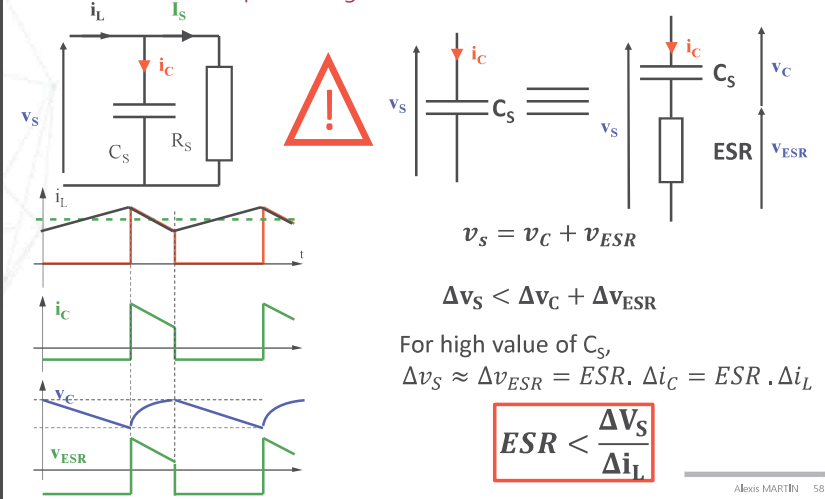
Inverters

Closed loop chopper control

Energy storage based chopper



Continuous mode: Output filtering



Alexis MARTIN 58

Introduction

Direct conversion

Buck chopper
Boost chopper

Reversible choppers

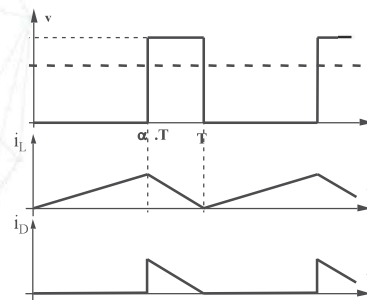
Inverters

Closed loop chopper control

Energy storage based chopper



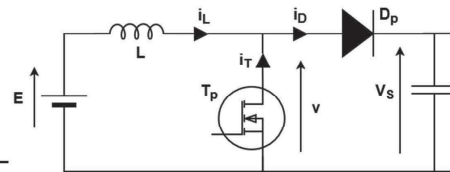
Boundary mode: $i_L(T)=0$



$$I_S = \langle i_D \rangle = \frac{1}{2} \cdot I_{L_{max}} \cdot (1-\alpha) = \frac{ET}{2L} \cdot \alpha(1-\alpha)$$

$$V_S = \frac{E}{1-\alpha}$$

$$I_S = \frac{ET}{2L} \cdot \frac{E}{V_S} \cdot \left(1 - \frac{E}{V_S}\right)$$



Alexis MARTIN 59

Introduction

Direct conversion

Buck chopper
Boost chopper

Reversible choppers

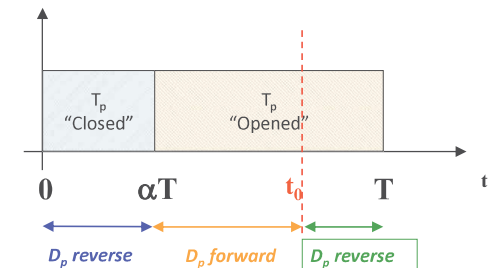
Inverters

Closed loop chopper control

Energy storage based chopper



Discontinuous mode



Alexis MARTIN 60

Introduction

Direct conversion

Buck chopper
Boost chopper

Reversible
choppers

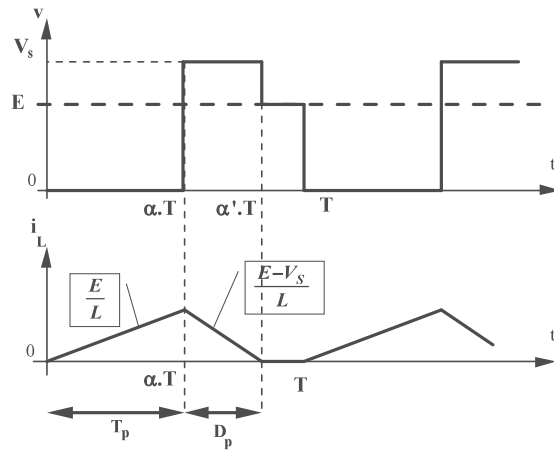
Inverters

Closed loop
chopper control

Energy storage
based chopper



Discontinuous mode: Chronograms



Alexis MARTIN 61

Introduction

Direct conversion

Buck chopper
Boost chopper

Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper



Discontinuous mode: calculation

$$\frac{E}{L} \alpha T = \frac{V_s - E}{L} (\alpha' - \alpha) T \Rightarrow V_s = \frac{\alpha'}{\alpha' - \alpha} E$$

$$I_s = \langle i_D \rangle = \frac{1}{2} \cdot I_{L \max} \cdot (\alpha' - \alpha) = \frac{ET}{2L} \cdot \alpha (\alpha' - \alpha)$$

$$\Rightarrow V_s = E + \frac{\alpha^2 E^2 T}{2LI_s}$$

Alexis MARTIN 62

Introduction

Direct conversion

Buck chopper
Boost chopper

Reversible
choppers

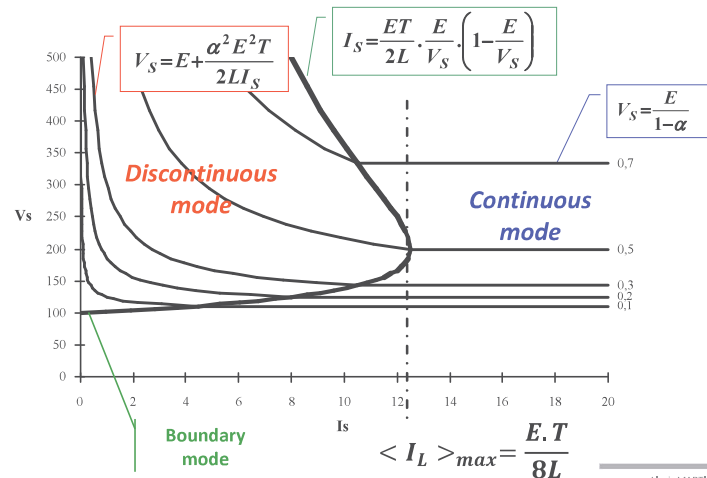
Inverters

Closed loop
chopper control

Energy storage
based chopper



Discontinuous mode: output characteristics



Alexis MARTIN 63

Introduction

Direct conversion

Buck chopper
Boost chopper

Reversible
choppers

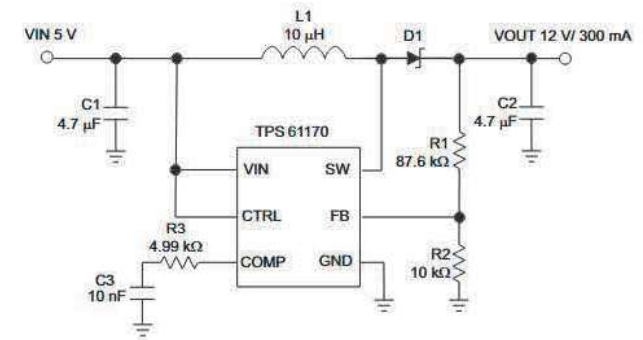
Inverters

Closed loop
chopper control

Energy storage
based chopper



Example : TPS61170



Alexis MARTIN 64

Introduction

Direct conversion

Buck chopper
Boost chopper

Reversible
choppers

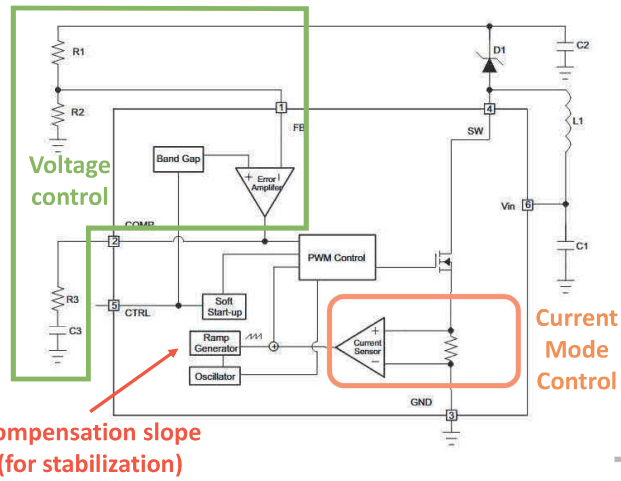
Inverters

Closed loop
chopper control

Energy storage
based chopper



Example : TPS61170



Alexis MARTIN 65

Introduction

Direct conversion

Buck chopper
Boost chopper

Reversible
choppers

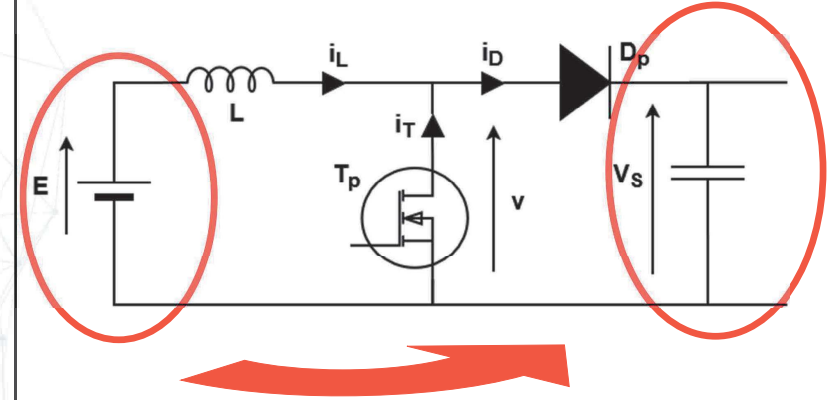
Inverters

Closed loop
chopper control

Energy storage
based chopper



Reminder : buck chopper



Alexis MARTIN 66

Introduction

Direct conversion

Buck chopper
Boost chopper

Reversible
choppers

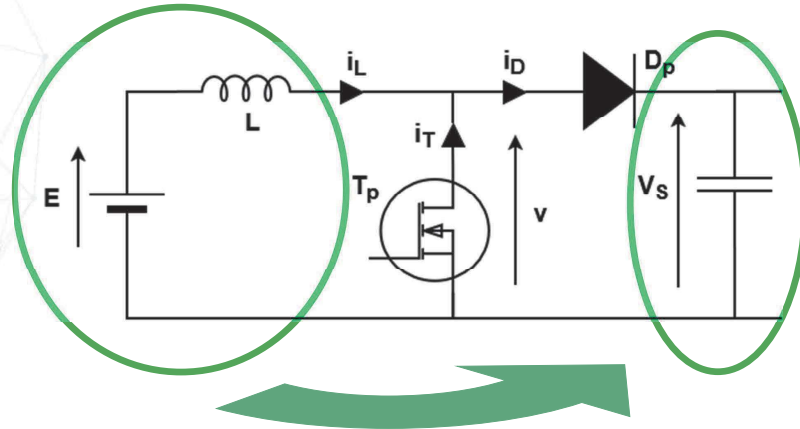
Inverters

Closed loop
chopper control

Energy storage
based chopper



Reminder : boost chopper



Alexis MARTIN 67

Introduction

Direct conversion

Buck chopper
Boost chopper

Reversible
choppers

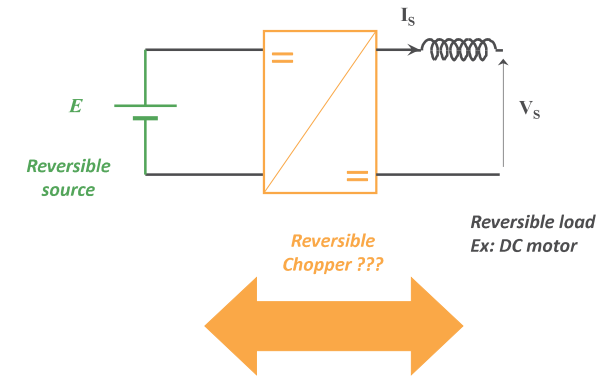
Inverters

Closed loop
chopper control

Energy storage
based chopper



Linked sources



Alexis MARTIN 68

Introduction

Direct conversion

Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper



Reversible Chopper

Introduction

Direct conversion

Reversible
choppers

2 quadrants chopper
4 quadrants chopper

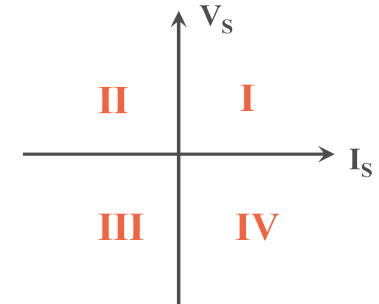
Inverters

Closed loop
chopper control

Energy storage
based chopper



Reversible choppers:
quadrants



Introduction

Direct conversion

Reversible
choppers
2 quadrants chopper
4 quadrants chopper

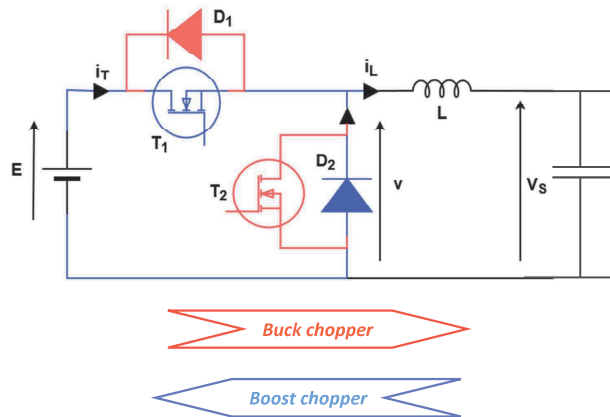
Inverters

Closed loop
chopper control

Energy storage
based chopper



Two quadrants reversible chopper



Introduction

Direct conversion

Reversible
choppers

2 quadrants chopper
4 quadrants chopper

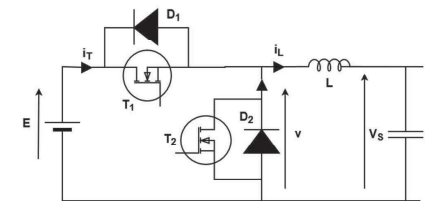
Inverters

Closed loop
chopper control

Energy storage
based chopper

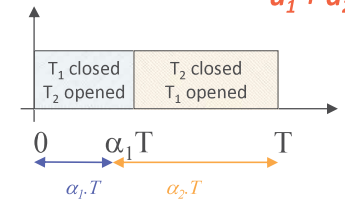


Two quadrants chopper:
control



2 switches => 2 duty cycles
Complementary control

$$\alpha_1 + \alpha_2 = 1$$



Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

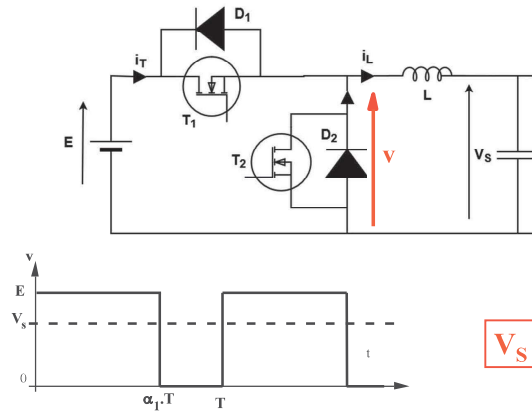
Inverters

Closed loop chopper control

Energy storage based chopper



Two quadrants chopper: output voltage



$$V_s = \alpha_1 \cdot E$$

Alexis MARTIN 73

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

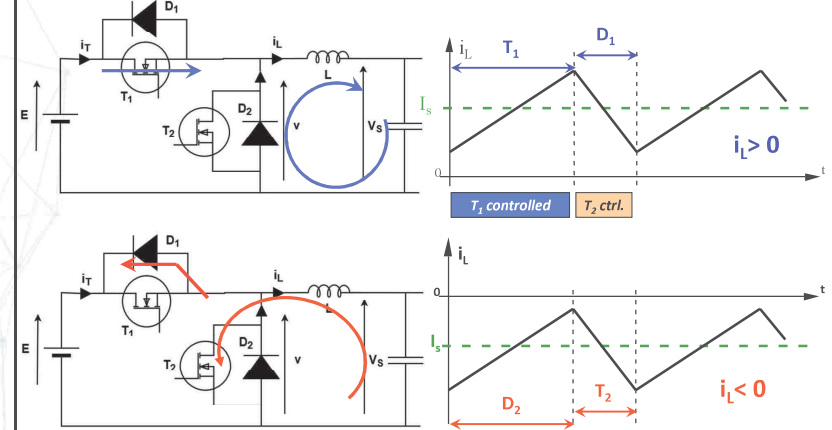
Inverters

Closed loop chopper control

Energy storage based chopper



Two quadrants chopper: output current



Alexis MARTIN 74

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

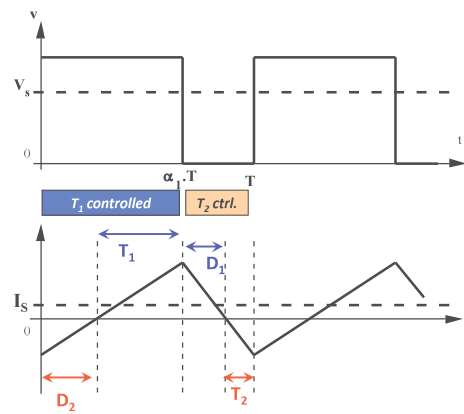
Inverters

Closed loop chopper control

Energy storage based chopper



Two quadrants chopper: Arbitrary output current



Alexis MARTIN 75

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

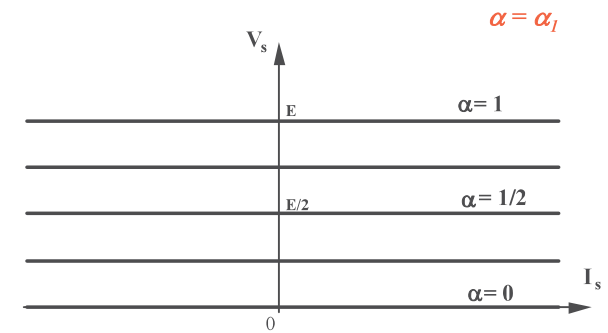
Inverters

Closed loop chopper control

Energy storage based chopper



Two quadrants chopper: Output characteristics grid



Alexis MARTIN 76

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

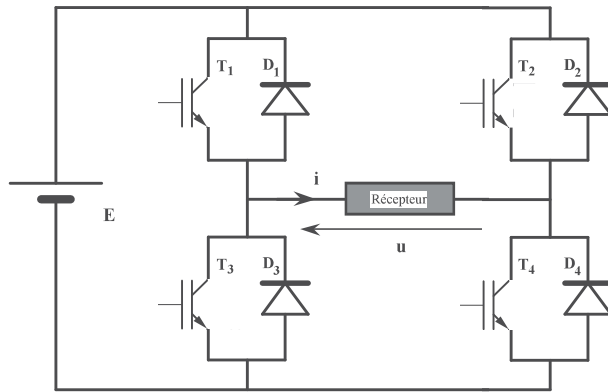
Inverters

Closed loop chopper control

Energy storage based chopper



4 quadrants chopper



Alexis MARTIN 77

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

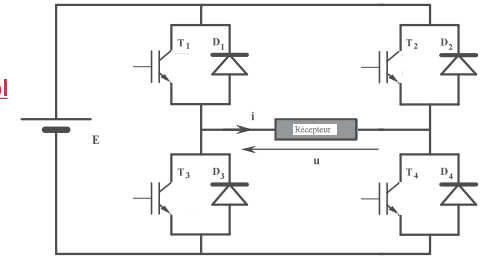
Inverters

Closed loop chopper control

Energy storage based chopper

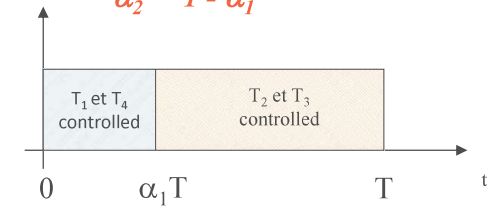


4 quadrants chopper:
Synchronous complementary control



$$\alpha_1 = \alpha_4 \text{ et } \alpha_2 = \alpha_3$$

$$\alpha_2 = 1 - \alpha_1$$



Alexis MARTIN 78

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

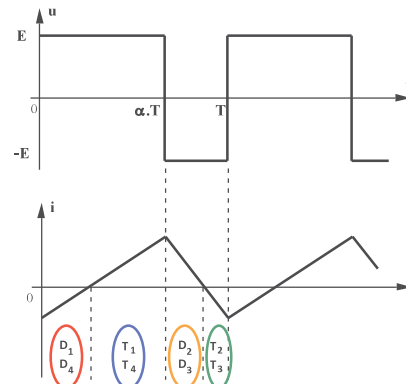
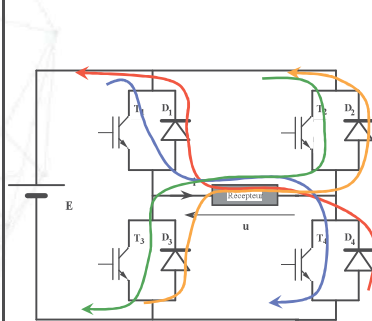
Inverters

Closed loop chopper control

Energy storage based chopper



4 quadrants chopper:
Synchronous complementary control (chronograms)



Alexis MARTIN 79

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

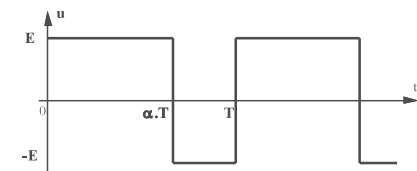
Inverters

Closed loop chopper control

Energy storage based chopper



4 quadrants chopper:
Synchronous complementary control (Average output voltage)



$$\langle u \rangle = (2\alpha - 1) \cdot E$$



High voltage ripple: $2 \cdot E$
Low switching frequency

Alexis MARTIN 80

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

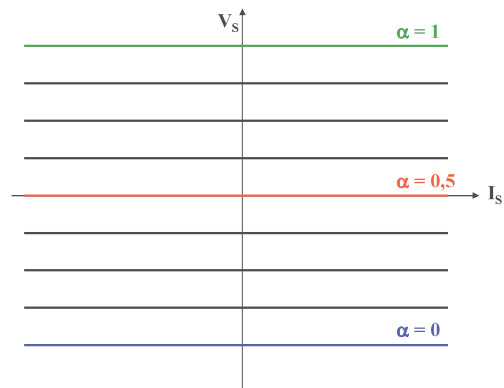
Inverters

Closed loop chopper control

Energy storage based chopper



4 quadrants chopper: Output characteristics



Alexis MARTIN B1

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

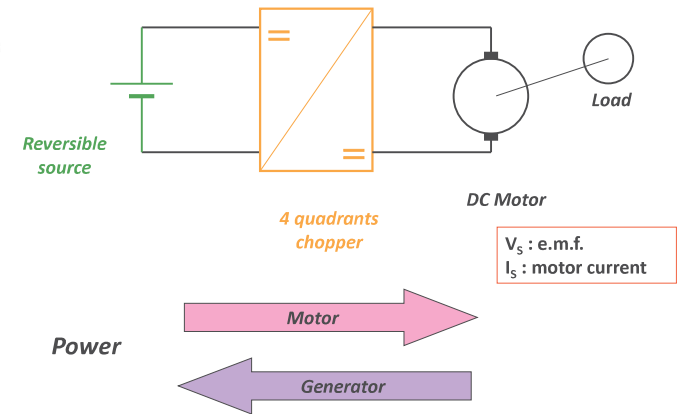
Inverters

Closed loop chopper control

Energy storage based chopper



4 quadrants chopper: control of a DC motor



Alexis MARTIN B2

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

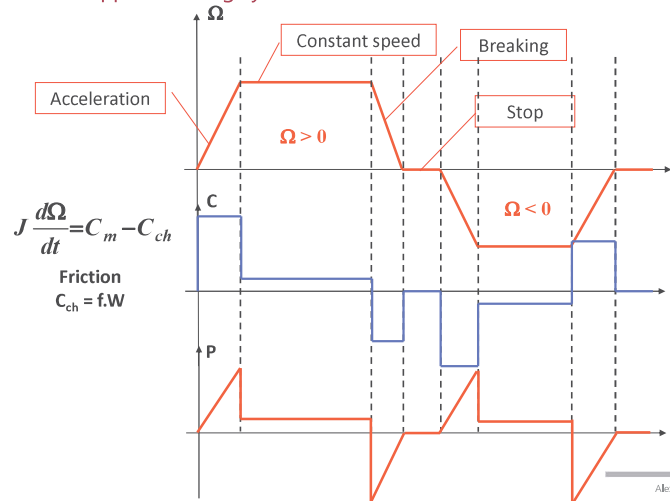
Inverters

Closed loop chopper control

Energy storage based chopper



4 quadrants chopper: working cycle



Alexis MARTIN B3

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

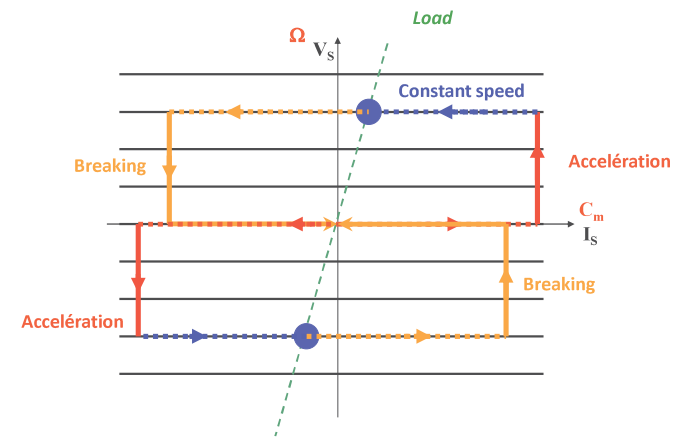
Inverters

Closed loop chopper control

Energy storage based chopper



4 quadrants chopper: Operating point



Alexis MARTIN B4

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

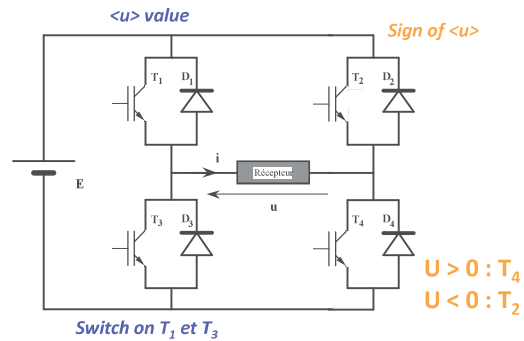
Inverters

Closed loop chopper control

Energy storage based chopper



4 quadrants chopper: additional control



Alexis MARTIN 85

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

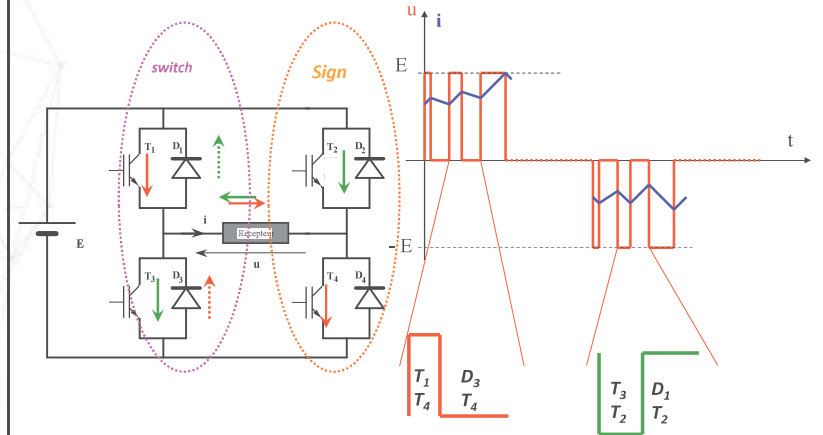
Inverters

Closed loop chopper control

Energy storage based chopper



4 quadrants chopper: additional control



Alexis MARTIN 86

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

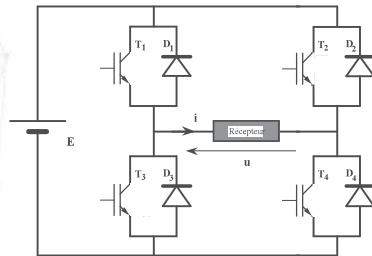
Inverters

Closed loop chopper control

Energy storage based chopper



4 quadrants chopper: additional control



$U = \pm a \cdot E$
Average value during
a switching time T_d



Voltage ripple reduced: E
Switching only on T_1/T_3 : losses reduced
Instabilities around 0V !!

Alexis MARTIN 87

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

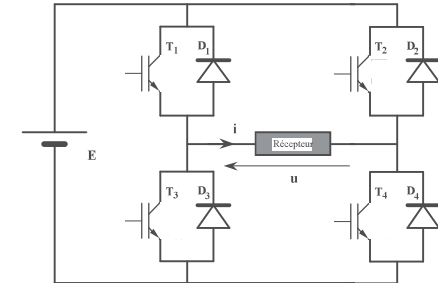
Inverters

Closed loop chopper control

Energy storage based chopper



4 quadrants chopper: Shifted complementary control



$$\alpha_1 = \alpha_1$$

$$\alpha_2 = \alpha_3$$

$$\alpha_3 = 1 - \alpha_1$$

$$\alpha_4 = 1 - \alpha_2$$

Switching on T_1 and T_3 at $t = 0$

Switching on T_2 and T_4 at $t = T/2$

Alexis MARTIN 88

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

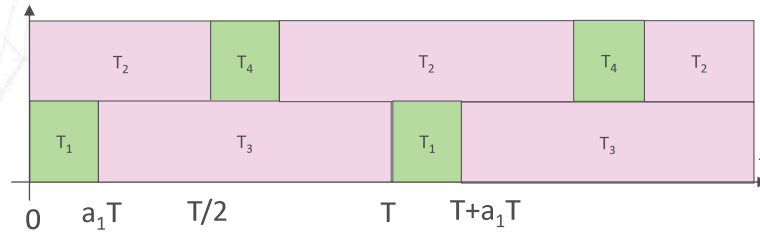
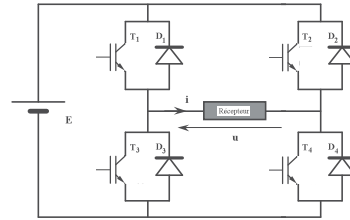
Inverters

Closed loop chopper control

Energy storage based chopper



4 quadrants chopper: Shifted complementary control



Alexis MARTIN 89

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

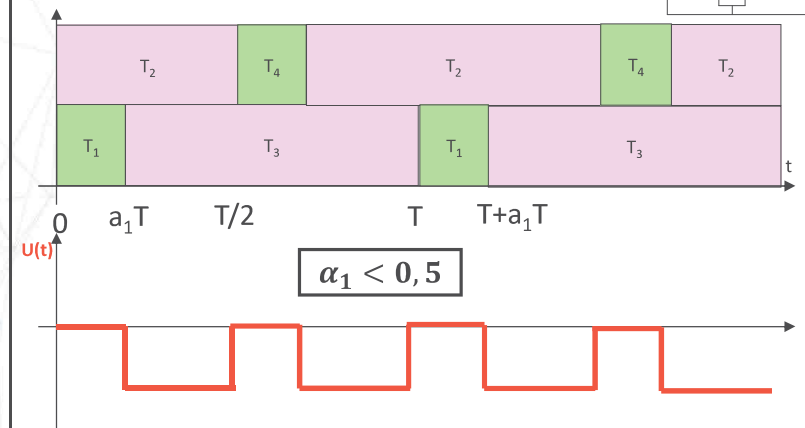
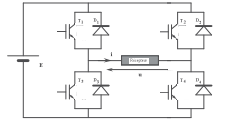
Inverters

Closed loop chopper control

Energy storage based chopper



4 quadrants chopper: Shifted complementary control



Alexis MARTIN 90

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

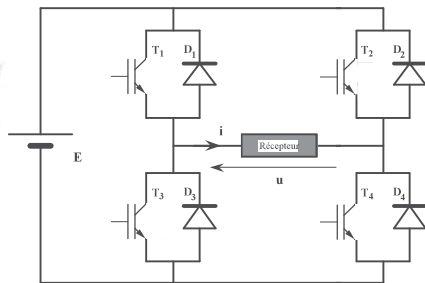
Inverters

Closed loop chopper control

Energy storage based chopper



4 quadrants chopper: Shifted complementary control



$$U = (2a - 1) \cdot E$$



Voltage ripple reduced: E
High switching frequency
=> PWM

Alexis MARTIN 91

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

Inverters

Closed loop chopper control

Energy storage based chopper

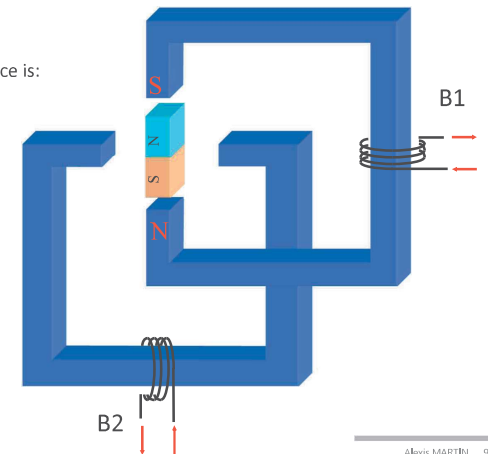


4 quadrants chopper: Stepper motor control

Operation principle: full step

For a clockwise rotation, the sequence is:

B1	B2
+I	0
0	+I
-I	0
0	-I



Alexis MARTIN 92

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

Inverters

Closed loop chopper control

Energy storage based chopper

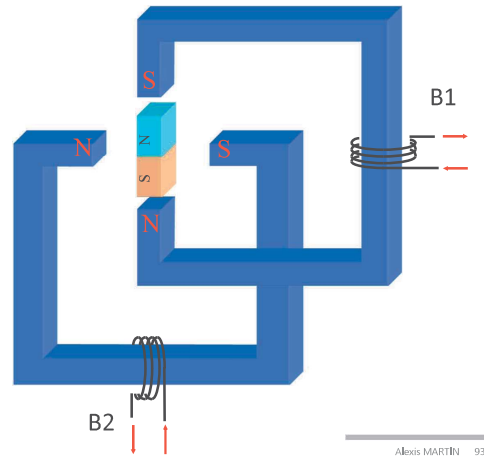


4 quadrants chopper: Stepper motor control

Operation principle: half step

For a clockwise rotation:

B1	B2
I+	0
I+	I+
0	I+
I-	I+
I-	0
I-	I-
0	I-
I+	I-



Alexis MARTIN 93

Introduction

Direct conversion

Reversible choppers

2 quadrants chopper
4 quadrants chopper

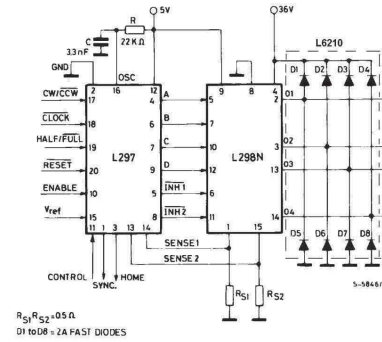
Inverters

Closed loop chopper control

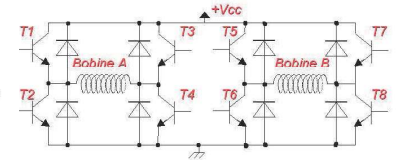
Energy storage based chopper



4 quadrants chopper: Stepper motor control



$R_{S1}, R_{S2} = 0.5 \Omega$
D1 to D8 = 2A FAST DIODES



Alexis MARTIN 94

Introduction

Direct conversion

Reversible choppers

Inverters

Closed loop chopper control

Energy storage based chopper



Inverter

Introduction

Direct conversion

Reversible choppers

Inverters

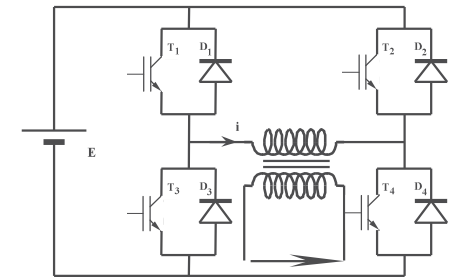
Single phase inverter
Three phase inverter
Three phase motor control

Closed loop chopper control

Energy storage based chopper



Single phase inverter



$$\alpha = \frac{1}{2} + \delta_a \sin(\omega t)$$

Alexis MARTIN 96

Introduction

Direct conversion

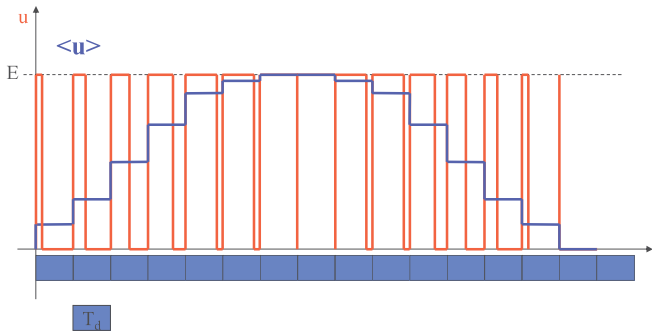
Reversible choppers

Inverters
Single phase inverter
Three phase inverter
Three phase motor control

Closed loop chopper control

Energy storage based chopper
ENSEA
Ecole Nationale Supérieure de l'Électronique et des Sciences Associées

Single phase inverter: PWM control



Introduction

Direct conversion

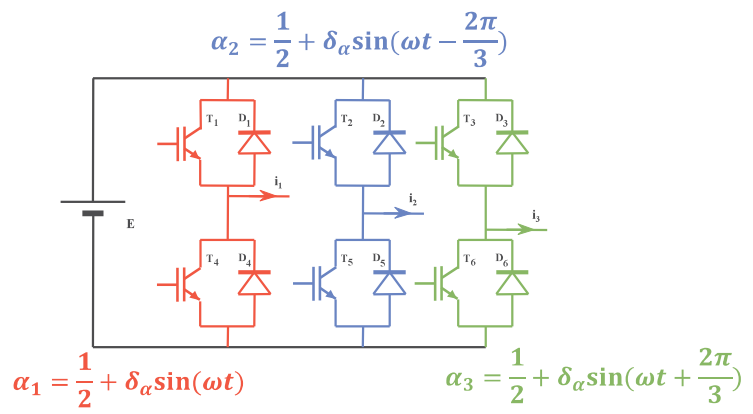
Reversible choppers

Inverters
Single phase inverter
Three phase inverter
Three phase motor control

Closed loop chopper control

Energy storage based chopper
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Three phase inverter



Introduction

Direct conversion

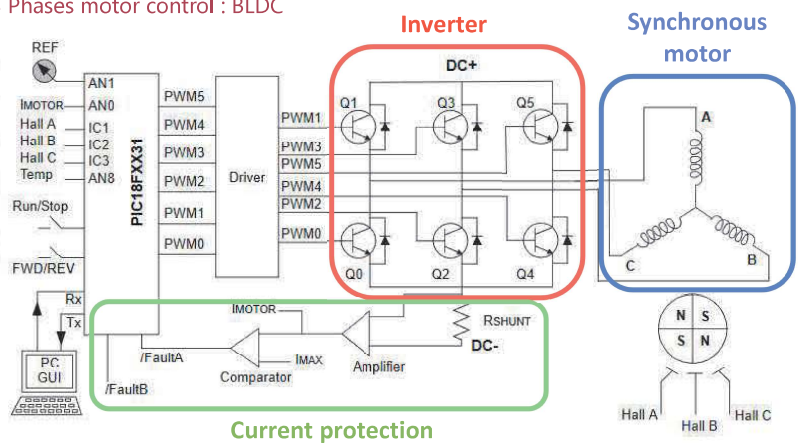
Reversible choppers

Inverters
Single phase inverter
Three phase inverter
Three phase motor control

Closed loop chopper control

Energy storage based chopper
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3 Phases motor control : BLDC



AN899 (Microchip) : Brushless DC Motor Control Using PIC18FXX31 MCUs

Introduction

Direct conversion

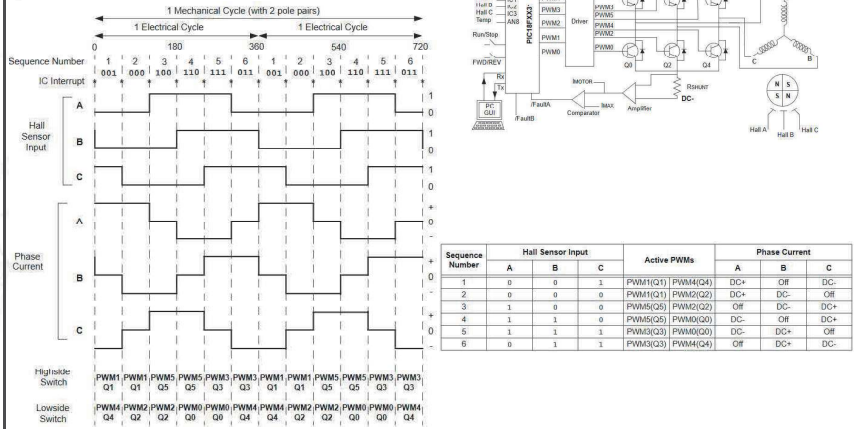
Reversible choppers

Inverters
Single phase inverter
Three phase inverter
Three phase motor control

Closed loop chopper control

Energy storage based chopper
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3 Phases motor control : BLDC



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Closed loop chopper control

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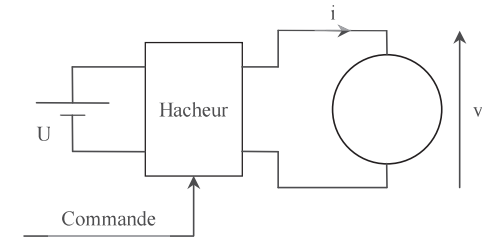
Closed loop
chopper control

Models
Current control
Speed control

Energy storage
based chopper



Closed loop chopper control



Possible control : speed / torque

Voltage / current

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DC motor model

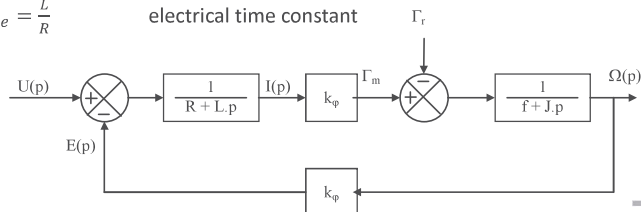
Electrical equation : $u = e + r \cdot i + L \cdot \frac{di}{dt}$

Mechanical equation : $J \frac{d\Omega}{dt} + f\Omega = \gamma_m - \gamma_r$

Coupling equations : $e = k_\phi \cdot \Omega \quad \gamma_m = k_\phi \cdot i$

$\tau_{em} = \frac{J \cdot R}{k_\phi^2 + R \cdot f}$ electromechanical time constant

$\tau_e = \frac{L}{R}$ electrical time constant



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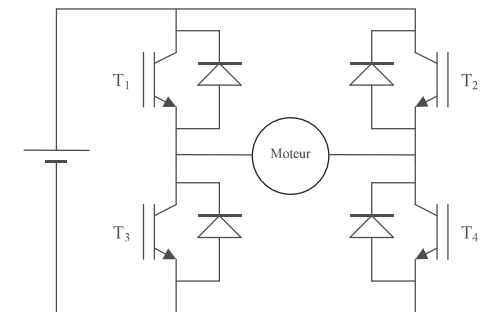
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Reversible structure



Chopper model

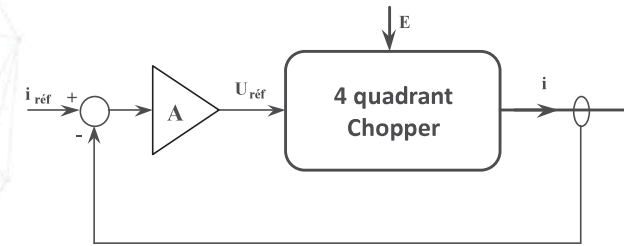
$$G_h(p) = \frac{U(p)}{\alpha(p)} = 2 \cdot V_{DC} \cdot e^{-\frac{T_d}{2} \cdot p}$$

(Average delay $T_d/2$)

$$G_h(p) \approx \frac{G_0}{1 + \tau_h \cdot p} \quad \text{with} \quad \tau_h = \frac{T_d}{2} \quad \text{and} \quad G_0 = 2 \cdot V_{DC}$$

(linearization)

Current control



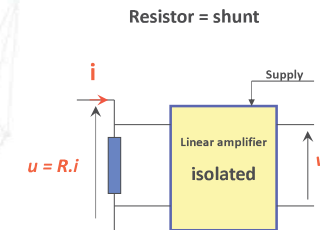
Torque control of a motor

Current loop

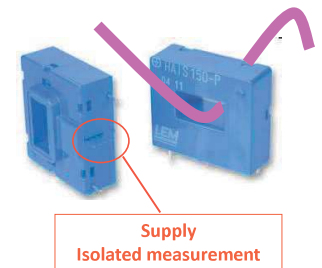
Armature current regulation

- Current regulation
- Torque control ($G = k_\phi \cdot I$)
- Better transient response
- Need a high performance sensor
- Simple Regulation (P.I.)

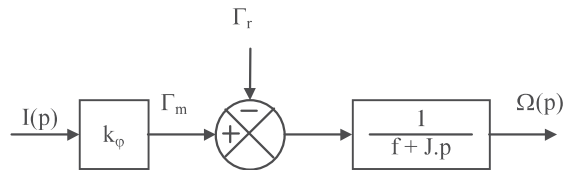
Current loop: Current sensor



Hall effect sensor

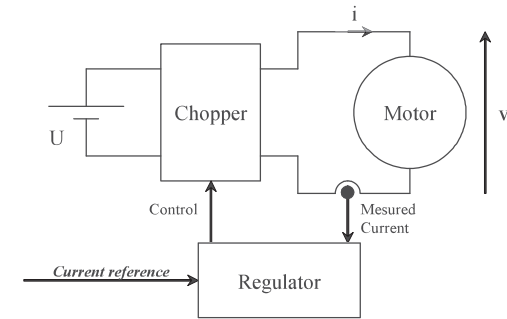


DC motor: Current control



$$G_i(p) = \frac{\Omega(p)}{I(p)} = \frac{k_\phi}{f + J.p}$$

Current loop



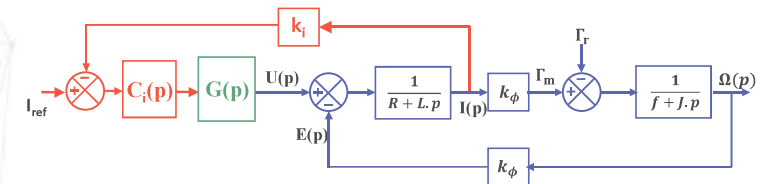
Control criteria

Accuracy : no static error

Speed : time response enhancement

Unconditional stability (high margin)

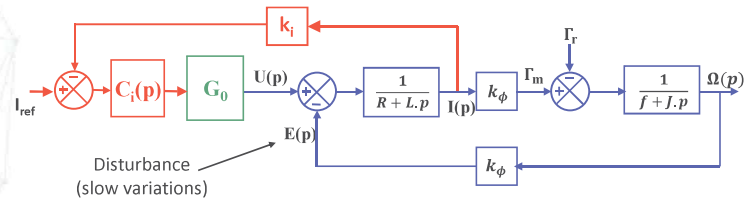
Current Control



Controller Chopper DC motor

= > Controller choice: $C_i(p)$

PI. control: simplified chopper



$$C_i(p) = k_i \cdot \frac{1 + \tau_i p}{\tau_i p}$$

$$H_{NCOL}(p) = \frac{G_0 / R}{1 + \tau_e p}$$

$$H_{OL}(p) = \frac{k_i \cdot (1 + \tau_i p) G_0 / R}{\tau_i p (1 + \tau_e p)}$$

$$\Rightarrow \tau_i = \tau_e$$

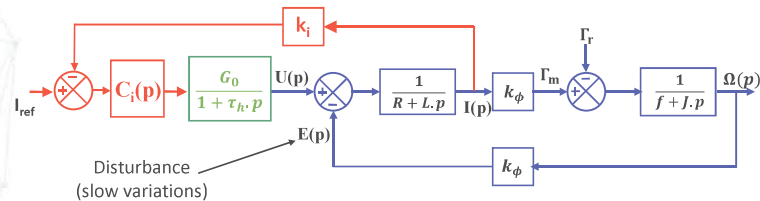
PI. control: simplified chopper

$$H_{CL}(p) = \frac{k_i \cdot G_0 / R}{k_i \cdot G_0 / R + \tau_e p} = \frac{1}{1 + \tau_{BF} p}$$

$$\tau_{BF} = \frac{\tau_e}{\frac{k_i \cdot G_0}{R}}$$

1st order without error, faster

PI. control: first order chopper



$$C_i(p) = k_i \cdot \frac{1 + \tau_i p}{\tau_i p}$$

$$H_{OL}(p) = \frac{G_0 / R}{(1 + \tau_h p) \cdot (1 + \tau_e p)}$$

$$H_{OL}(p) = \frac{k_i \cdot (1 + \tau_i p) G_0 / R}{\tau_i p (1 + \tau_h p) \cdot (1 + \tau_e p)}$$

$$\Rightarrow \tau_i = \max(\tau_h, \tau_e)$$

Closed loop

$$H_{OL}(p) = \frac{k_i G_0 / R}{\tau_{he} p (1 + \tau_{he} p)}$$

$$H_{CL}(p) = \frac{1}{1 + 2m \frac{p}{\omega_0} + \frac{p^2}{\omega_0^2}}$$

*Second order without error**Choose of m : damping [0,7 .. 1]**Choose of ω_0 : bandwidth*

Introduction

Direct conversion

Reversible
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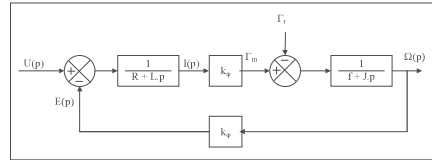
Models
Current control
Speed control

Energy storage
based chopper



DC motor: Speed / voltage transfer function

$$G'_{\omega}(p) = \frac{\Omega(p)}{U(p)} = \frac{k_{\phi}}{k_{\phi}^2 + (R + L.p).(f + J.p)}$$



$$G_u(p) = \frac{\Omega(p)}{U(p)} = \frac{k_{\phi}}{k_{\phi}^2 + Rf} \cdot \frac{1}{(1 + \tau_e p).(1 + \tau_{em} p)}$$

$$\tau_{em} = \frac{J.R}{k_{\phi}^2 + Rf} \quad \text{electromechanical time constant}$$

$$\tau_e = \frac{L}{R} \quad \text{electrical time constant}$$

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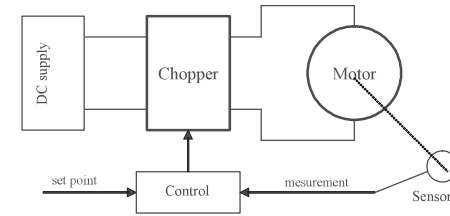
Closed loop
chopper control

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Speed loop



$$G_{hv}(p) = \frac{V_{\Omega_{mes}}(p)}{V_{\Omega_{ref}}(p)} = \frac{G_0 \Omega}{1 + 2m \frac{p}{\omega_0} + \frac{p^2}{\omega_0^2}}$$

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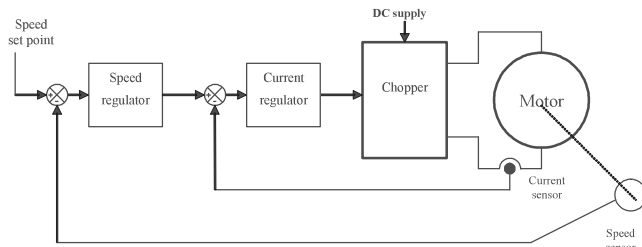
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Nested loop



Internal loop faster than external loop

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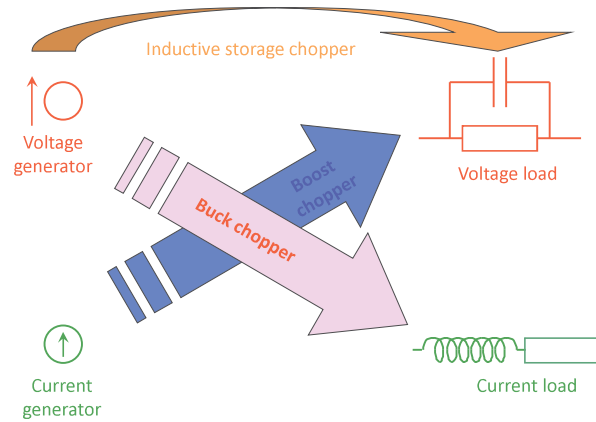
Closed loop
chopper control

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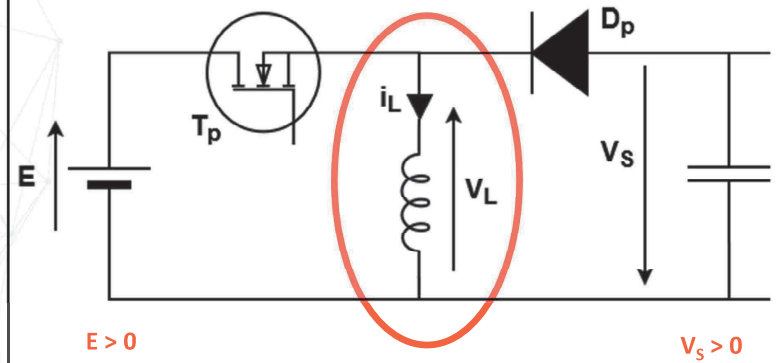
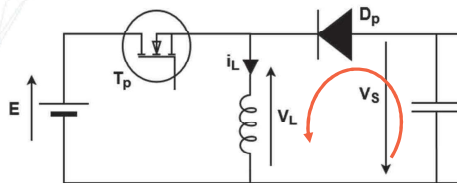
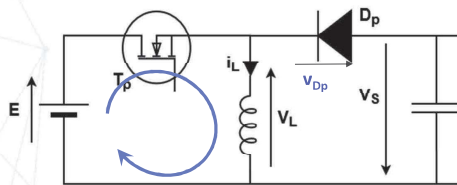


Indirect Choppers

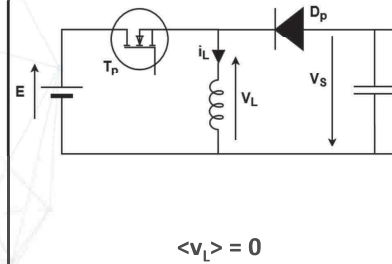
Indirect chopper: principle



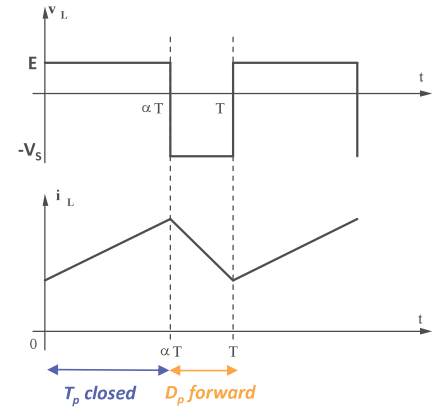
Buck-Boost structure

Buck-boost: Continuous mode ($i_L > 0$)

Buck-boost: Continuous mode (chronograms)



Inductive storage chopper



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Energy storage based chopper

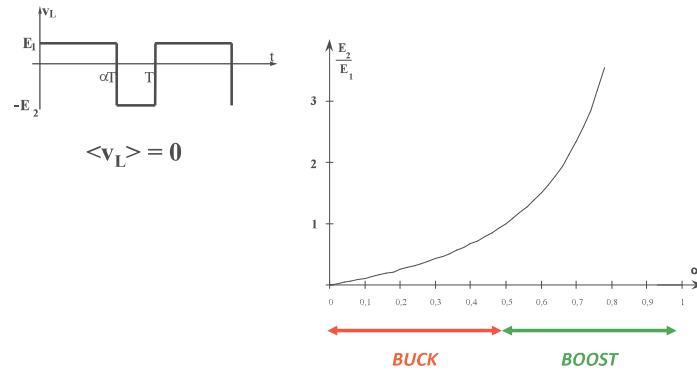
Buck-Boost

Cuk

SEPIC



Buck-boost: Continuous mode (Transfer characteristics)



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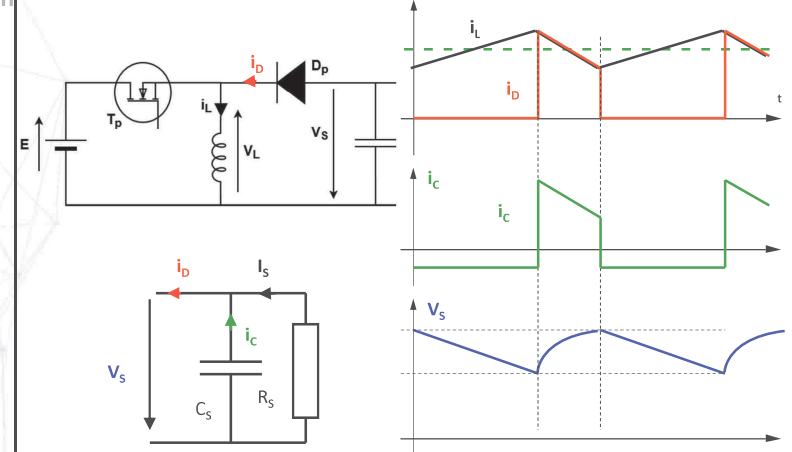
Buck-Boost

Cuk

SEPIC



Buck-boost: Continuous mode (output filtering)



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Energy storage based chopper

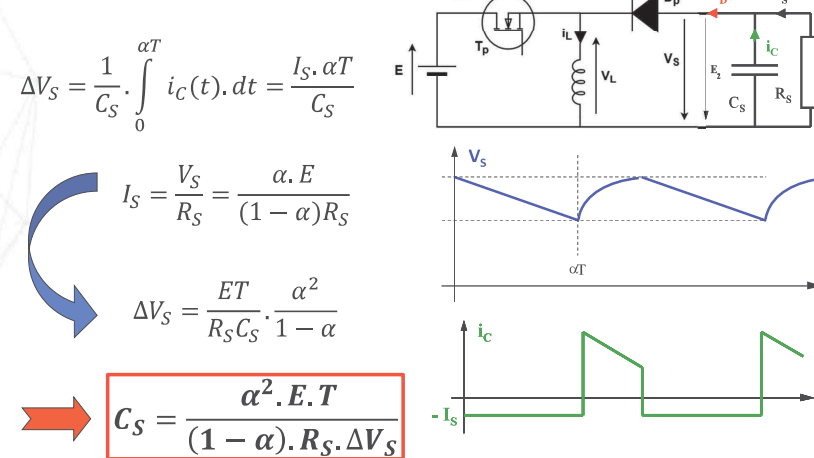
Buck-Boost

Cuk

SEPIC



Buck-boost: Continuous mode (Output filtering)



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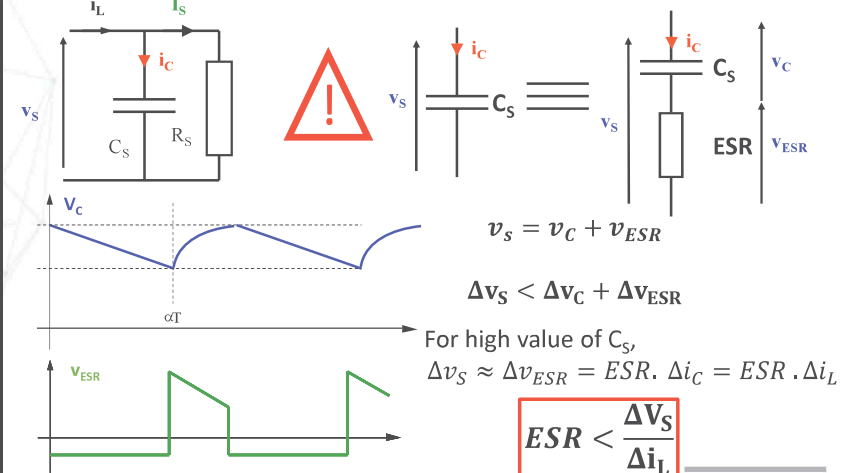
Buck-Boost

Cuk

SEPIC



Buck-boost: Continuous mode (Output filtering)



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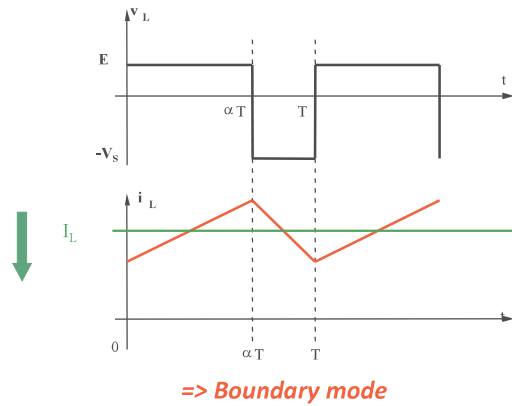
Buck-Boost

Cuk

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Buck-boost: "decreasing" load



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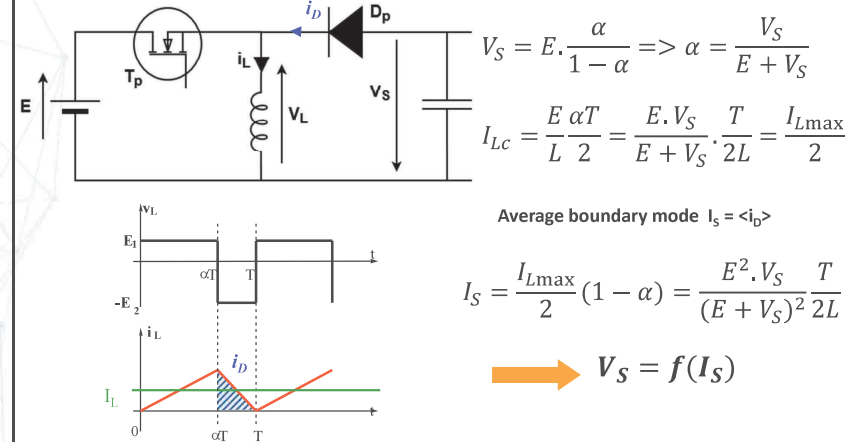
Buck-Boost

Cuk

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Buck-boost: Boundary mode



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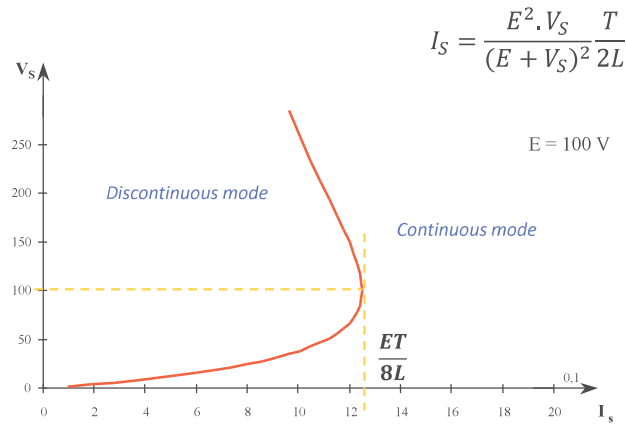
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Buck-boost: Boundary mode (characteristics)



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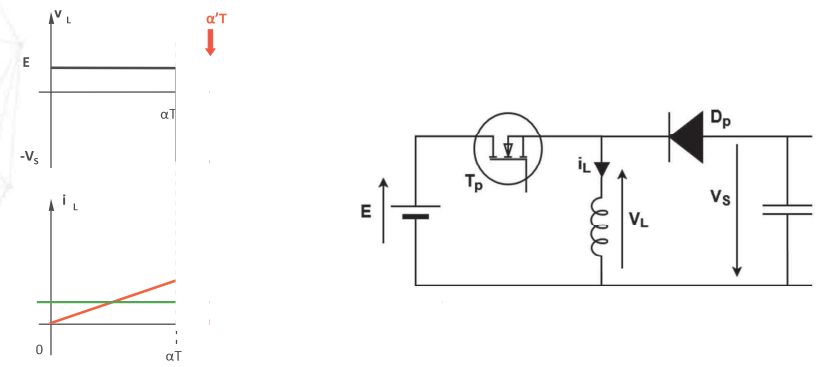
Buck-Boost

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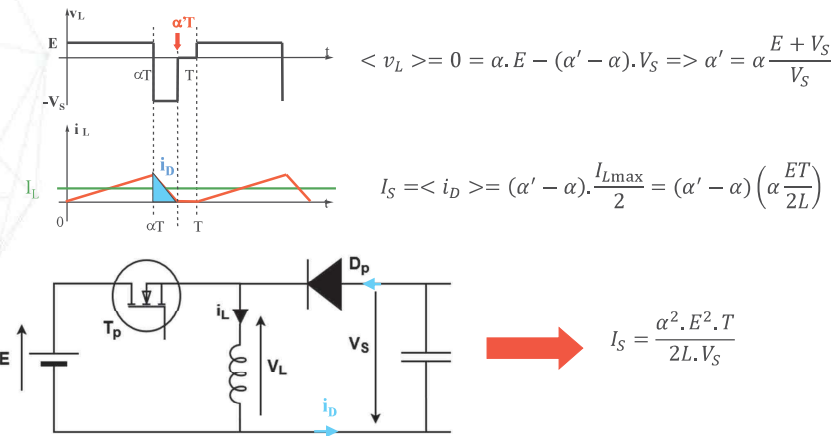
Buck-boost: Discontinuous mode



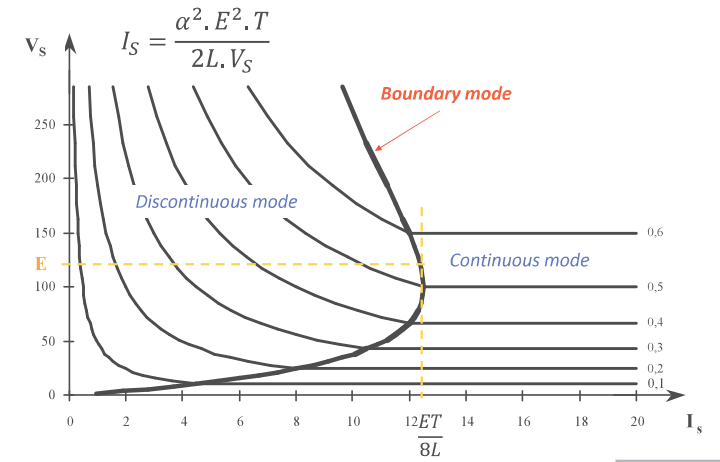
Full study: Switched-mode power supply

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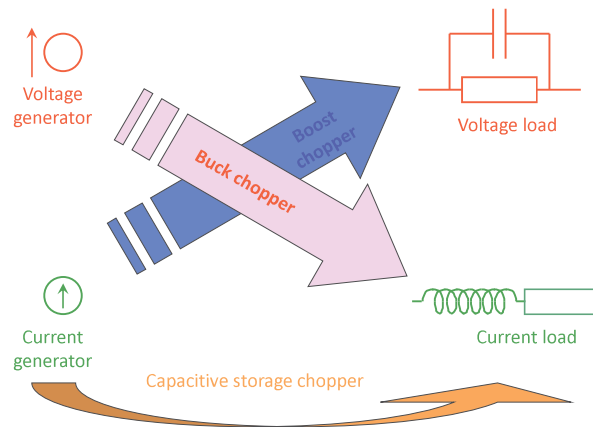
Buck-boost: Discontinuous mode



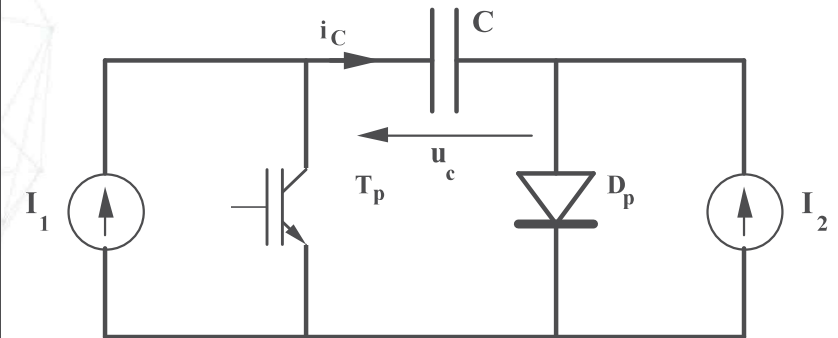
Buck-boost: Output characteristics



Capacitive storage chopper



Cuk structure



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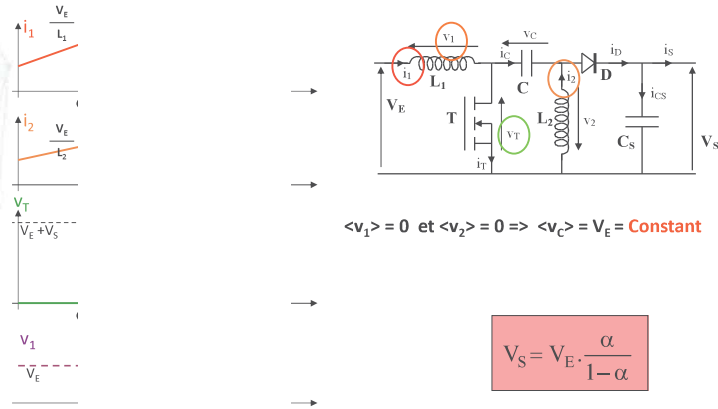
Buck-Boost

Cuk

SEPIC



SEPIC: Continuous mode (voltage transfer)



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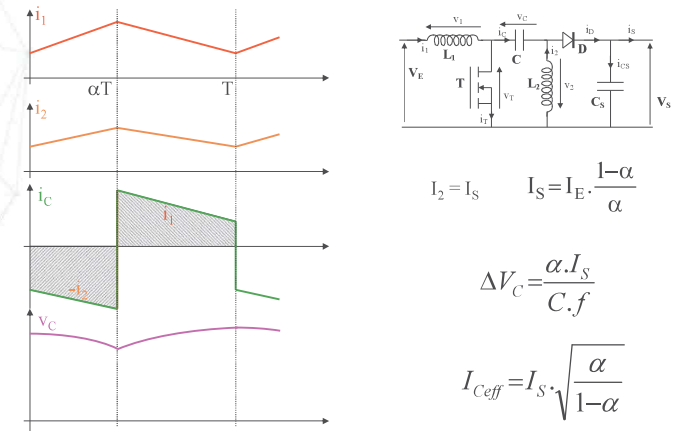
Energy storage based chopper

Buck-Boost

Cuk



SEPIC: Continuous mode (coupling capacitor)



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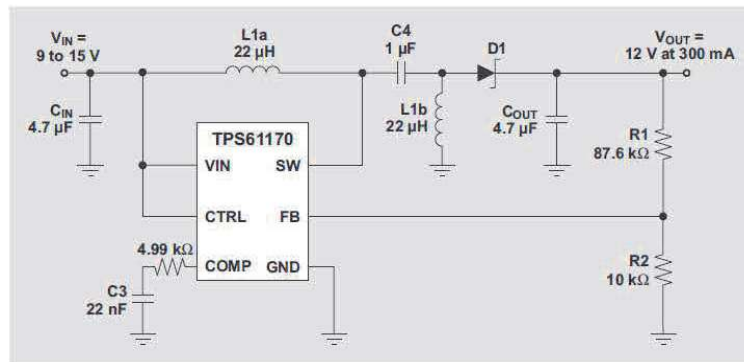
Buck-Boost

Cuk

SEPIC



SEPIC: circuit example TPS 61170 – Texas Instrument



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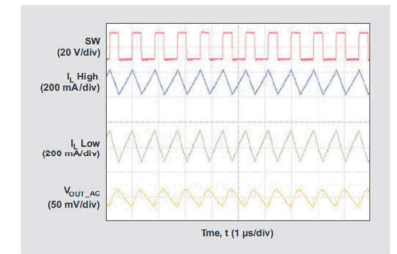
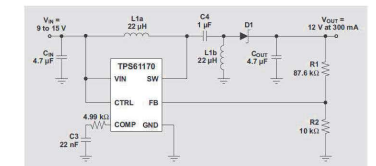
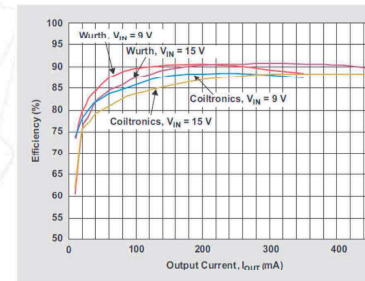
Energy storage based chopper

Buck-Boost

Cuk



SEPIC: circuit example TPS 61170 – Texas Instrument



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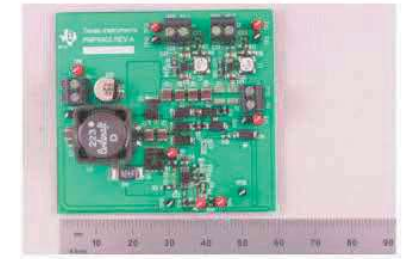


SEPIC
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Beyond Engineering

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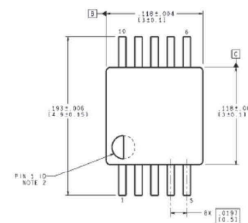
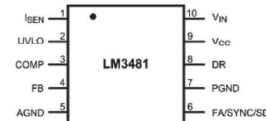
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